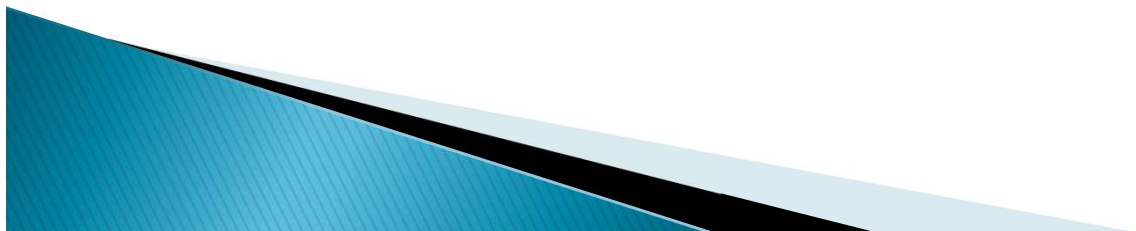


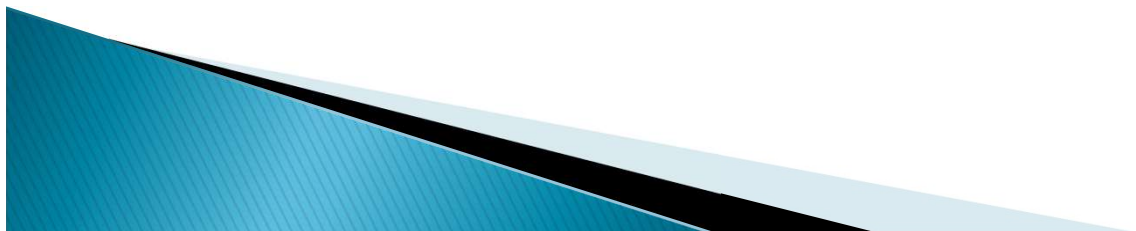
If $\mathbf{B}$ is added to $\mathbf{C} = (3.0 \text{ m}) \hat{i} + (4.0 \text{ m}) \hat{j}$, the result is a vector in the positive direction of the y-axis, with a magnitude equal to that of $\mathbf{C}$. What is the magnitude of $\mathbf{B}$?



If $\mathbf{B}$ is added to $\mathbf{C} = (3.0 \text{ m}) \hat{i} + (4.0 \text{ m}) \hat{j}$, the result is a vector in the positive direction of the y-axis, with a magnitude equal to that of $\mathbf{C}$. What is the magnitude of $\mathbf{B}$?

STRATEGY:

**Sketch, indicate given information,
Apply Definitions,
Simplify**



Chapter 3 – Vectors

OBJECTIVES: DEFINITIONS & APPLICATIONS

Transform between Polar & Cartesian systems,

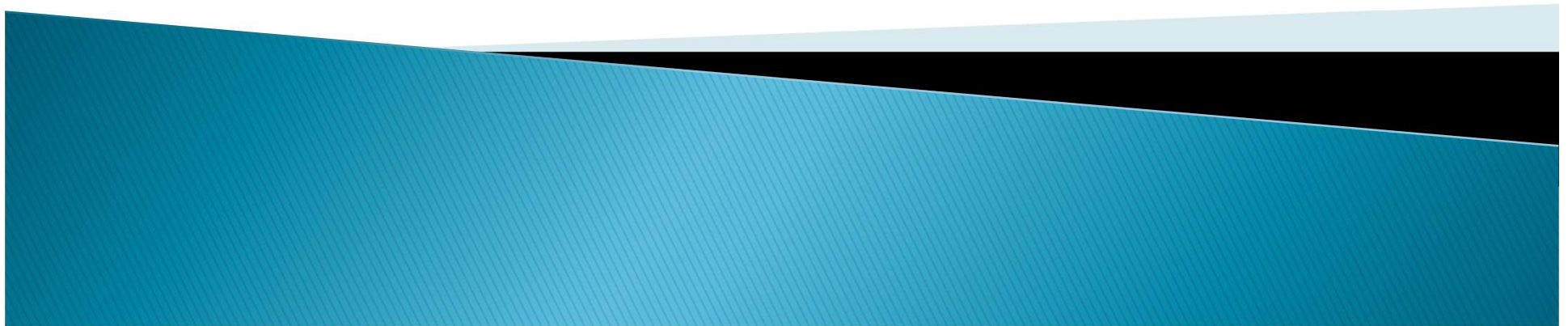
Properties of vectors

Differentiate between Vectors and Scalars

Components of vectors

Unit vectors

Determining resultants using unit vectors



Coordinate System

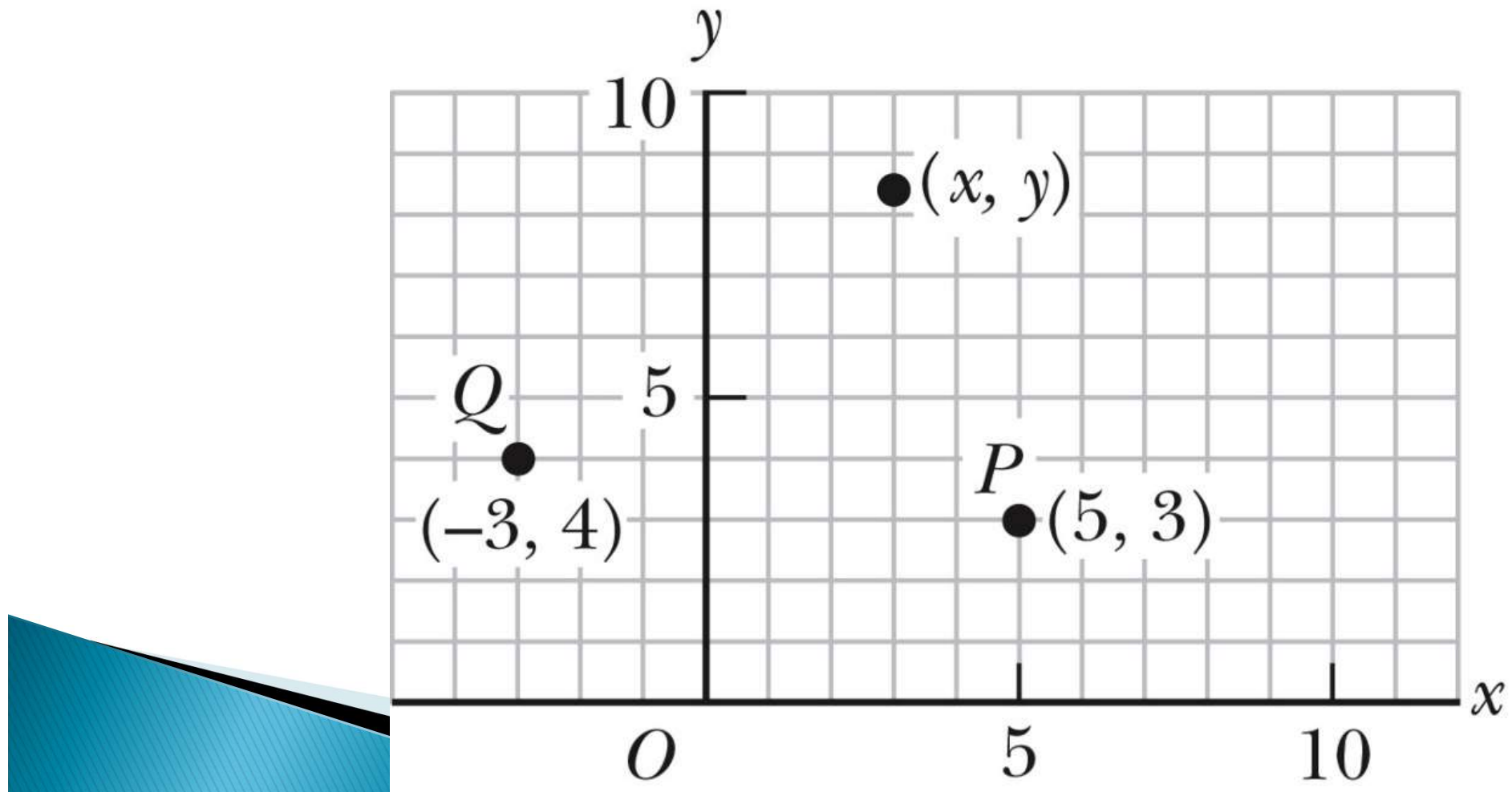
- ▶ Physics often requires a **description of a location in space**
 - e.g. an object's motion as in previous chapter 2.
- ▶ **Use various ways to represent this location**
 - Cartesian or Rectangular coordinates
 - Plane polar coordinates
- ▶ Need to apply trigonometric functions



Cartesian or Rectangular Coordinate System

x - and y - axes intersect at the origin

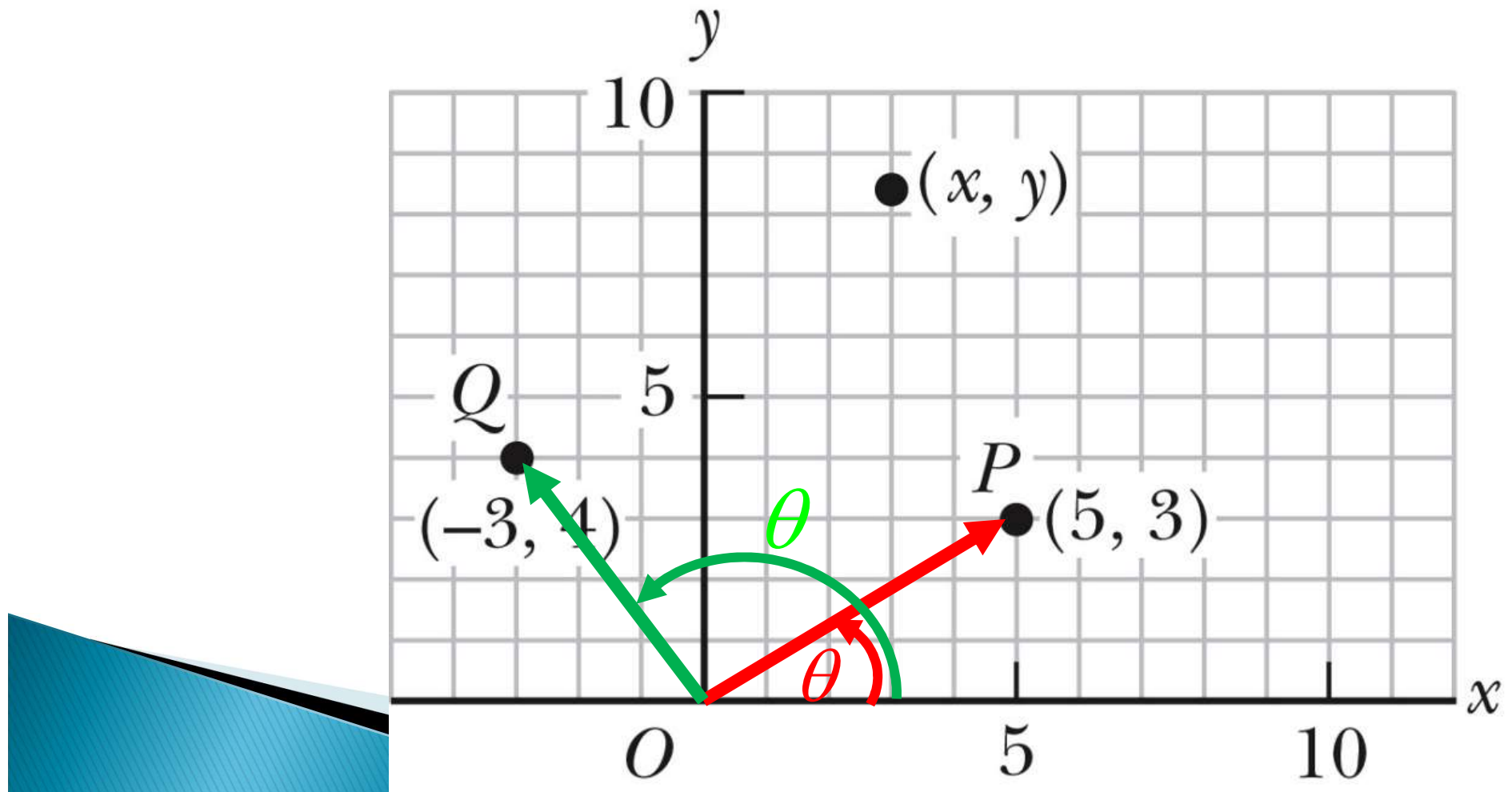
Points are labeled (x, y)



Cartesian or Rectangular Coordinate System

x - and y - axes intersect at the origin

Points are labeled (x, y)

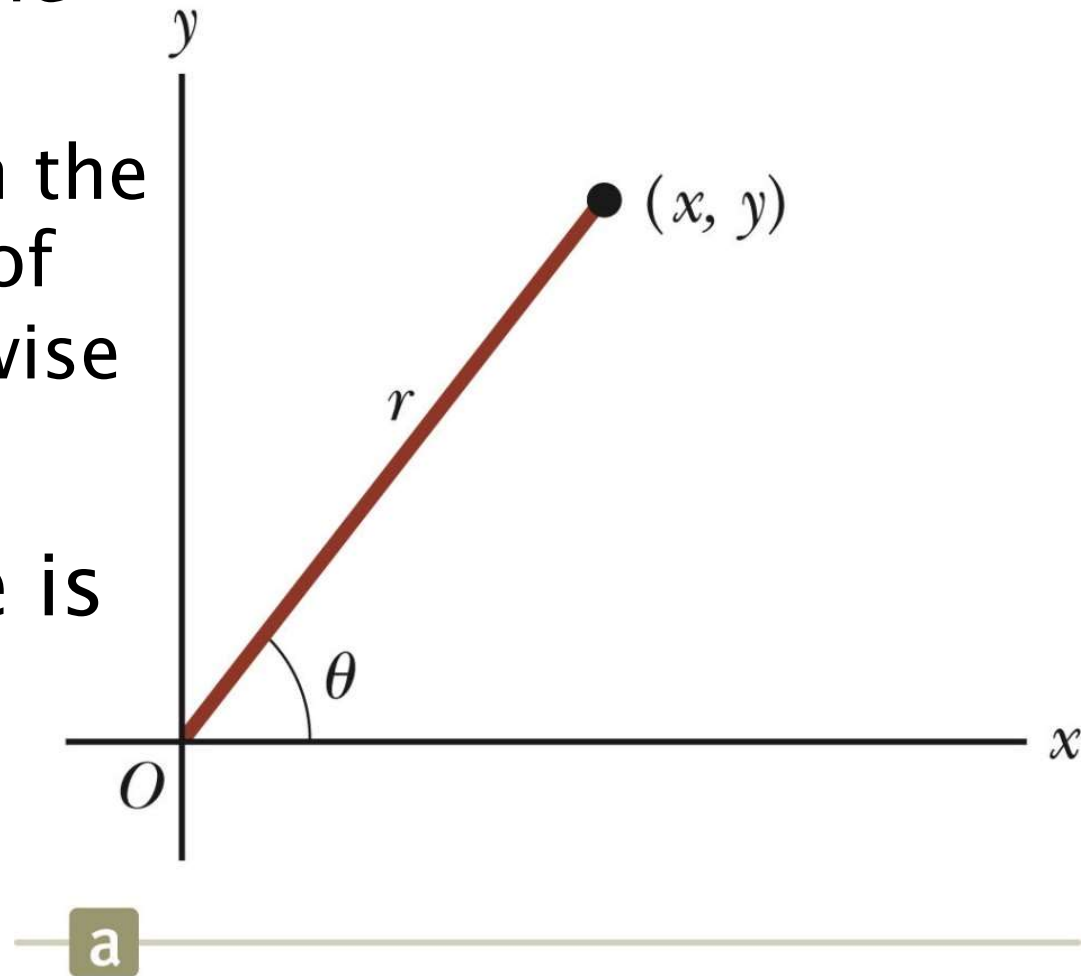


Polar Coordinate System

Origin and reference line are noted

Point is distance r from the origin in the direction of angle θ , counter clockwise from reference line

- The reference line is often the x -axis.
- Points are labeled (r, θ)



Polar $\leftrightarrow$ Cartesian Coordinates

Based on forming a right triangle from r and θ

If the Polar coordinates are known:

$$x = r \cos \theta$$

$$y = r \sin \theta$$

If the Cartesian coordinates are known:

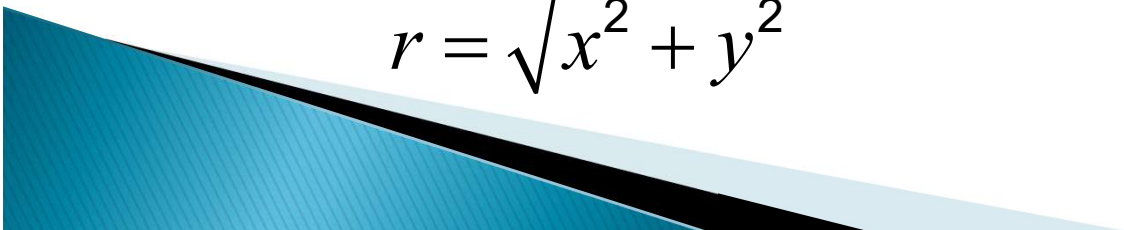
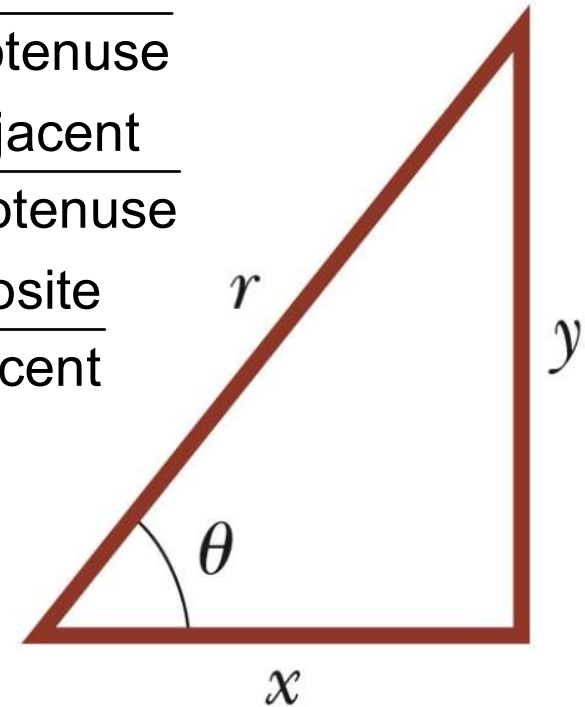
$$\tan \theta = \frac{\text{opp}}{\text{adj}} = y/x$$

$$r = \sqrt{x^2 + y^2}$$

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$



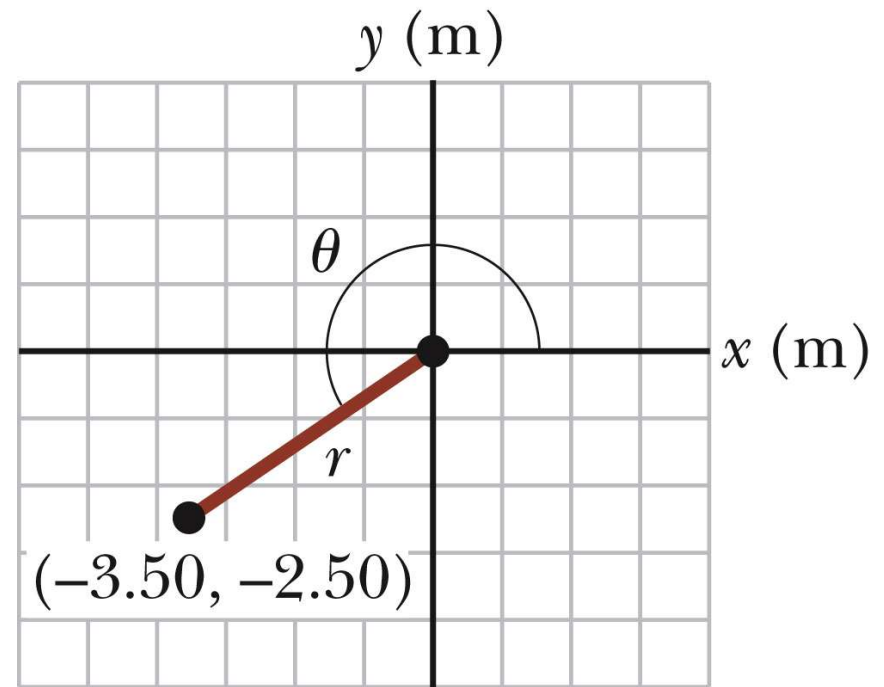
Example 3.1

The Cartesian coordinates of a point in the x - y plane are $(x, y) = (-3.50, -2.50)$ m, as shown in the figure. Find the polar coordinates of this point.

$$r = \sqrt{x^2 + y^2} = \dots = 4.30 \text{ m}$$

$$\theta = \tan^{-1}\left(\frac{\text{opp}}{\text{adj}}\right) = \tan^{-1}\left(\frac{y}{x}\right) = \dots = 216^\circ$$

- ▶ Repeat with the reference line along the minus x -axis



Scalars and Vectors

- ▶ A *scalar* quantity is completely described by a single value with an appropriate unit and has no direction.
 - temperature, speed, mass

- ▶ A *vector* is a physical quantity and is completely specified by a number with an appropriate unit (magnitude) plus a direction:

- velocity, force, displacement

- $\vec{F}$ or $\vec{F}$ or F or $\vec{F}$ in texts

- Handwritten: $\vec{F}$

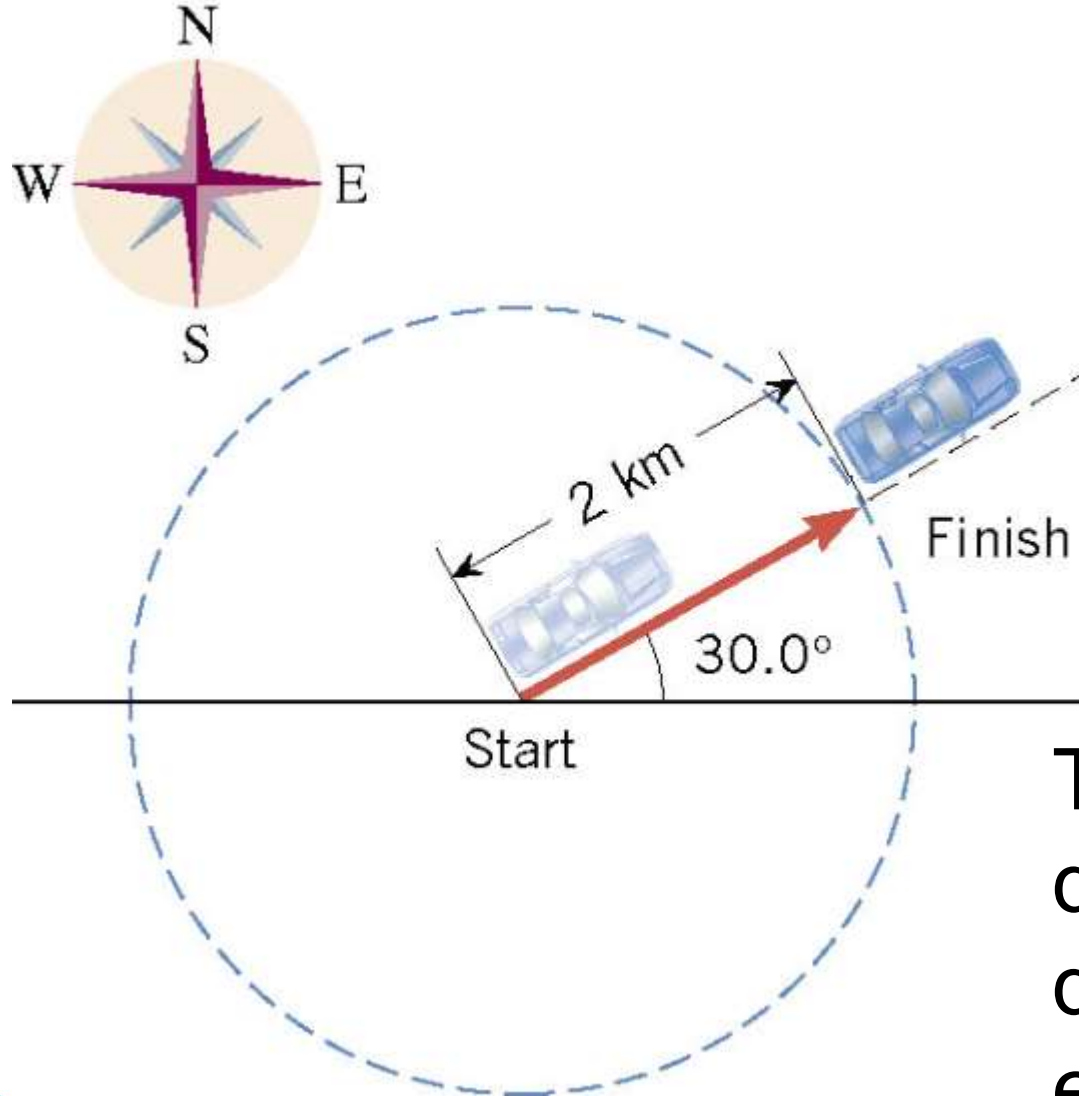
$\vec{F}$ or $\vec{F}$ or F or $\vec{F}$

- Magnitude of the vector:

$$|\vec{F}|$$

$$\therefore F = |\vec{F}|$$

Scalars vs Vectors



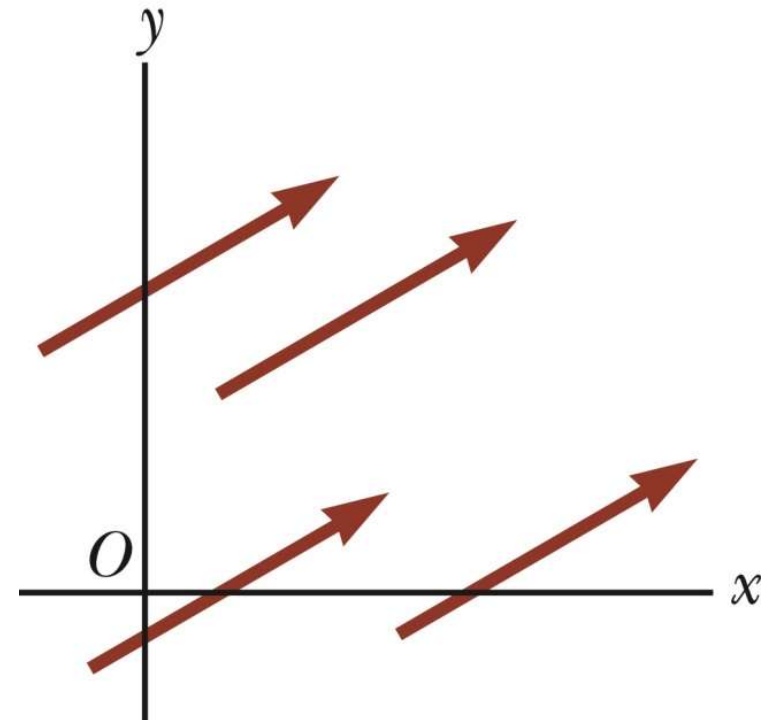
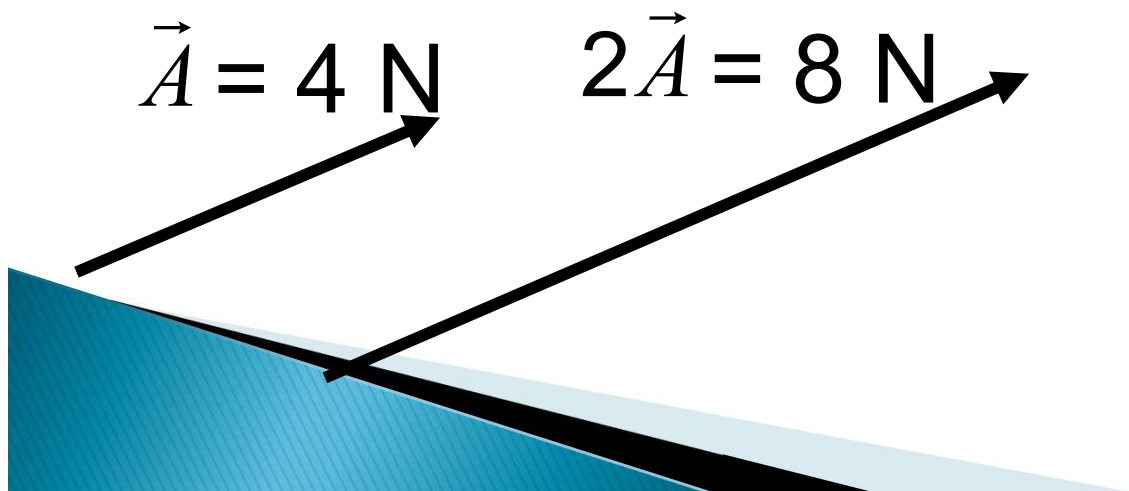
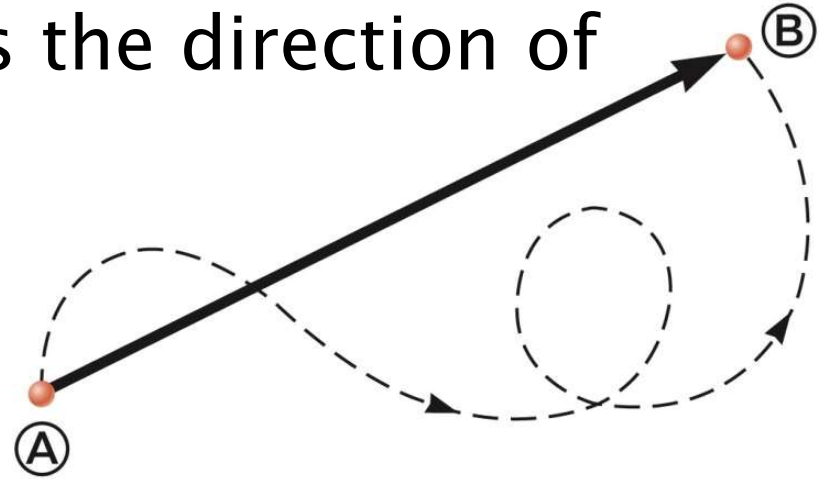
The car moved a distance of 2 km.

The car moved a distance of 2 km in a direction 30° north of east.

Vectors

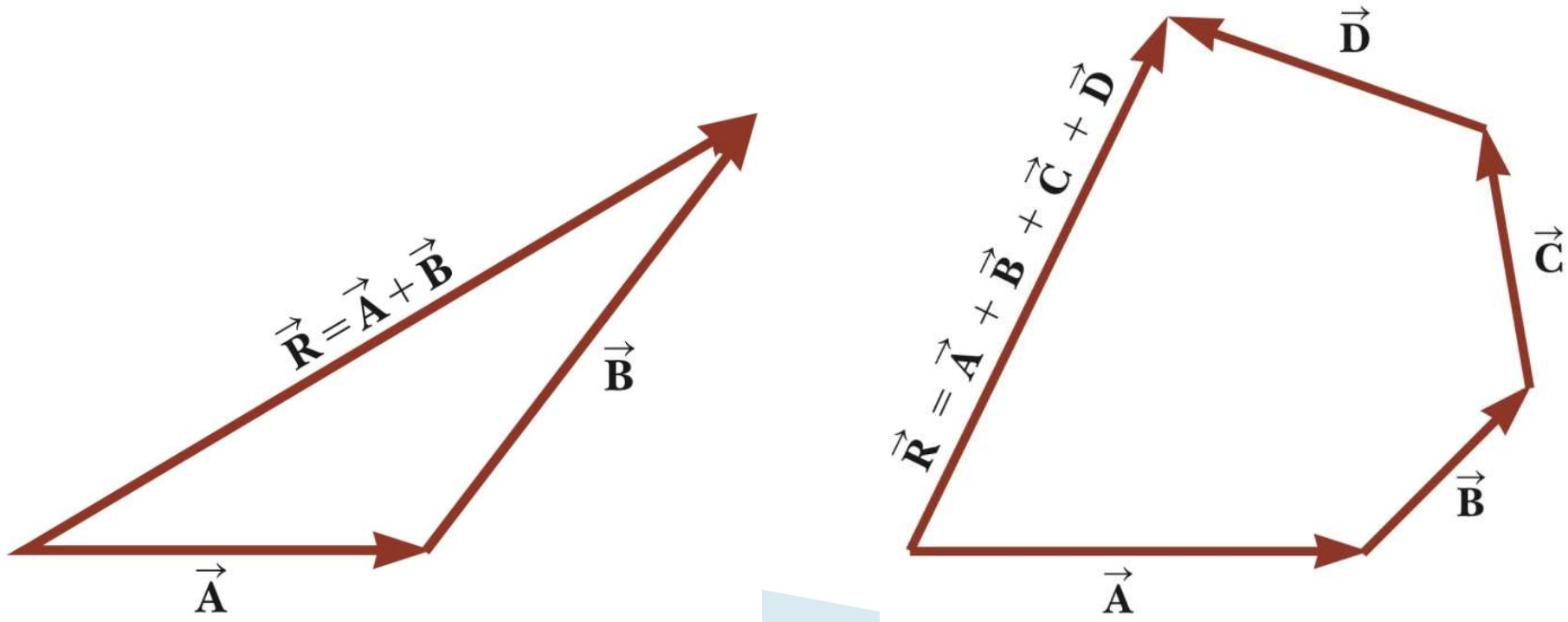
- ▶ Arrows are used to represent vectors. The direction of the arrow gives the direction of the vector.

- ▶ By convention, the length of a vector arrow is proportional to the magnitude of the vector.



Vector addition–Geometric

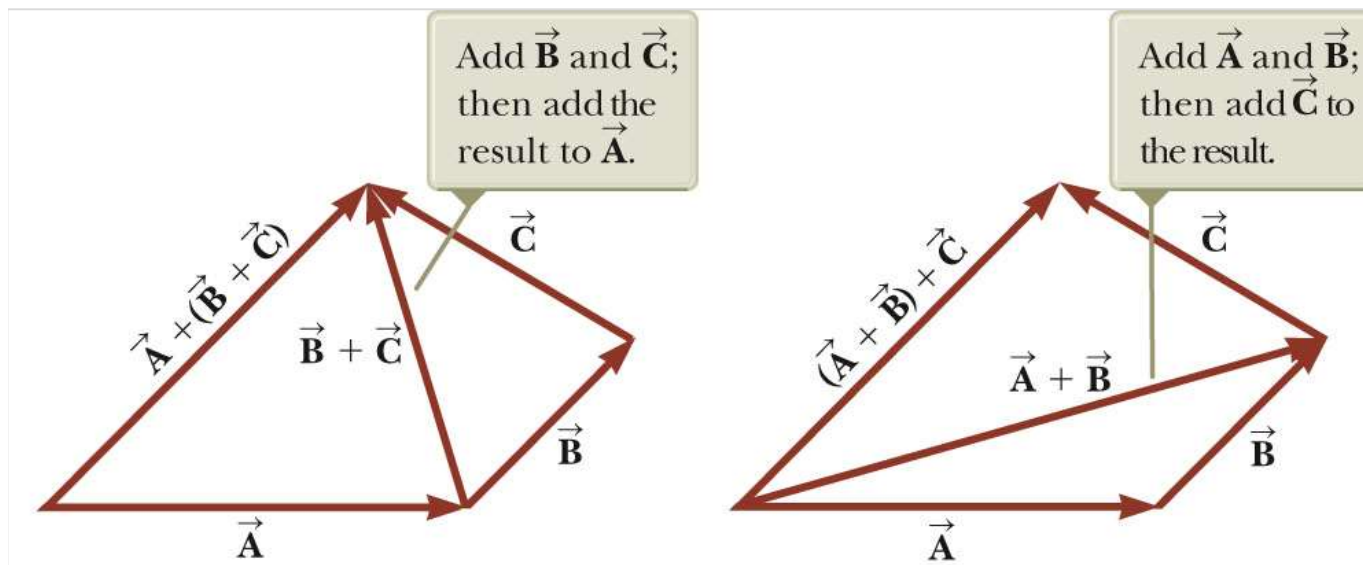
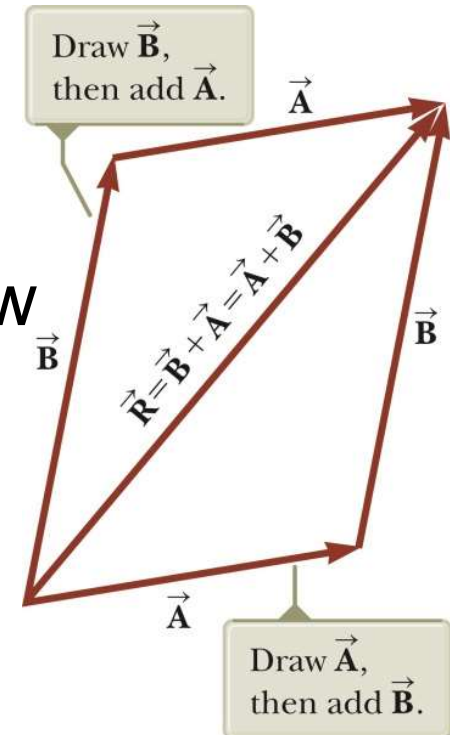
- ▶ A particle moves from point 1 to point 2, then from point 2 to point 3.
- ▶ Draw first vector to scale, then the second vector at the tip of the first.
- ▶ This is called: “Head to Tail” method.
- ▶ Resultant drawn: Starting point to End point



Vector addition–Geometric

- ▶ Commutative law **Holds**
- ▶ Associative law **Holds**

▶ Commutative law



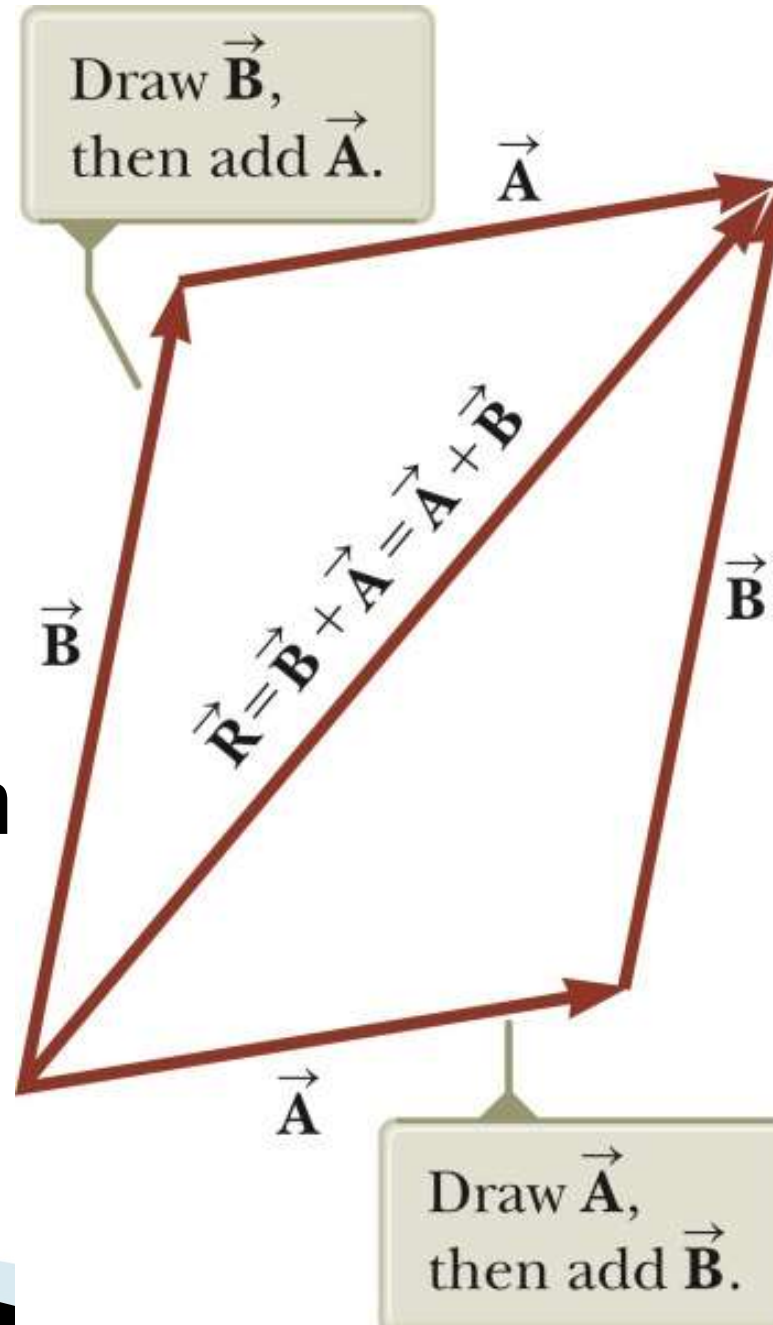
$$\vec{A} + (\vec{B} + \vec{C}) = (\vec{A} + \vec{B}) + \vec{C} = \vec{A} + \vec{B} + \vec{C}$$

▶ Associative law

Vector addition–Geometric

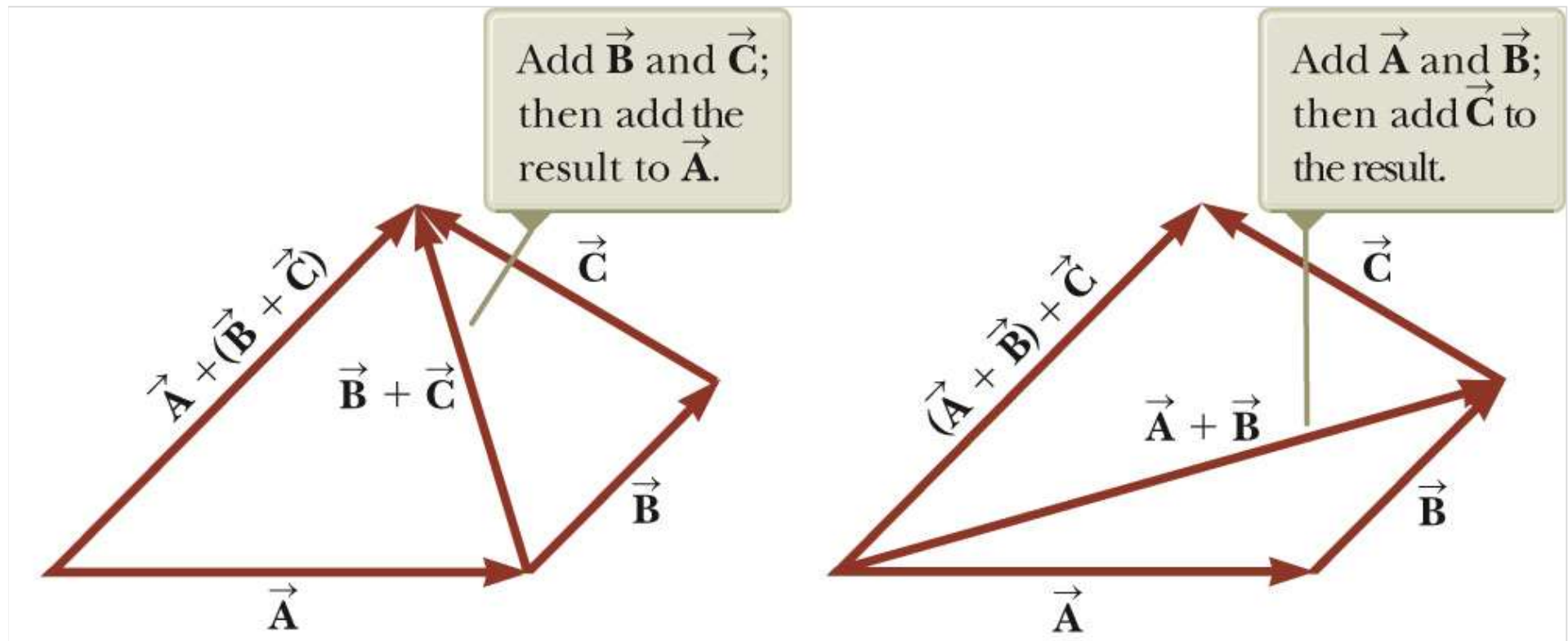
- ▶ Commutative law
- ▶ Associative law

Parallelogram
method



Vector addition–Geometric

- ▶ Commutative law
- ▶ Associative law

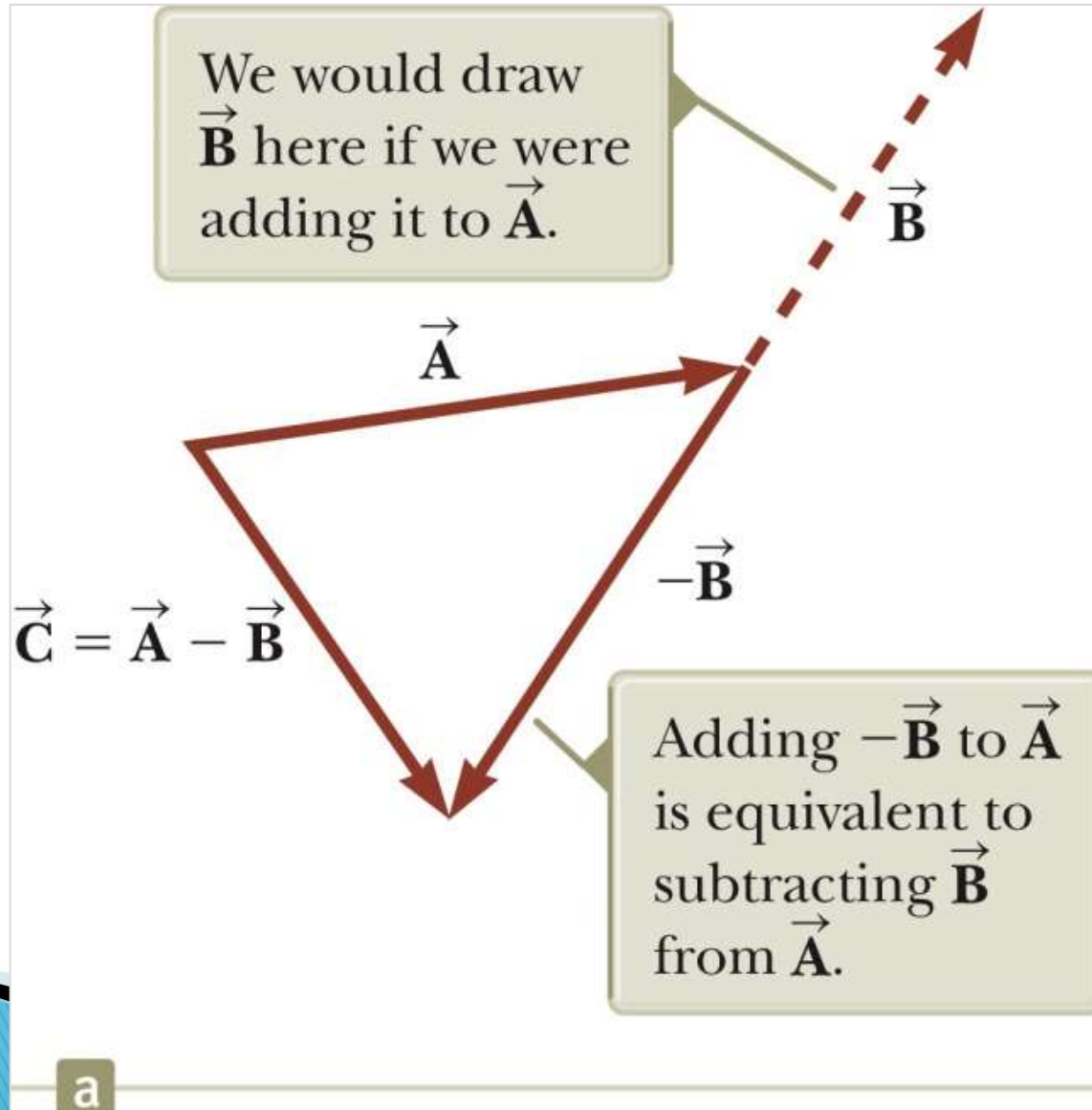


$$\vec{A} + (\vec{B} + \vec{C}) = (\vec{A} + \vec{B}) + \vec{C} \quad \text{Associative Law}$$

Vector addition–Geometric

- ▶ Vector subtraction
(Negative vector)

$$\vec{A} + (-\vec{B}) = \vec{C} = \vec{A} - \vec{B}$$

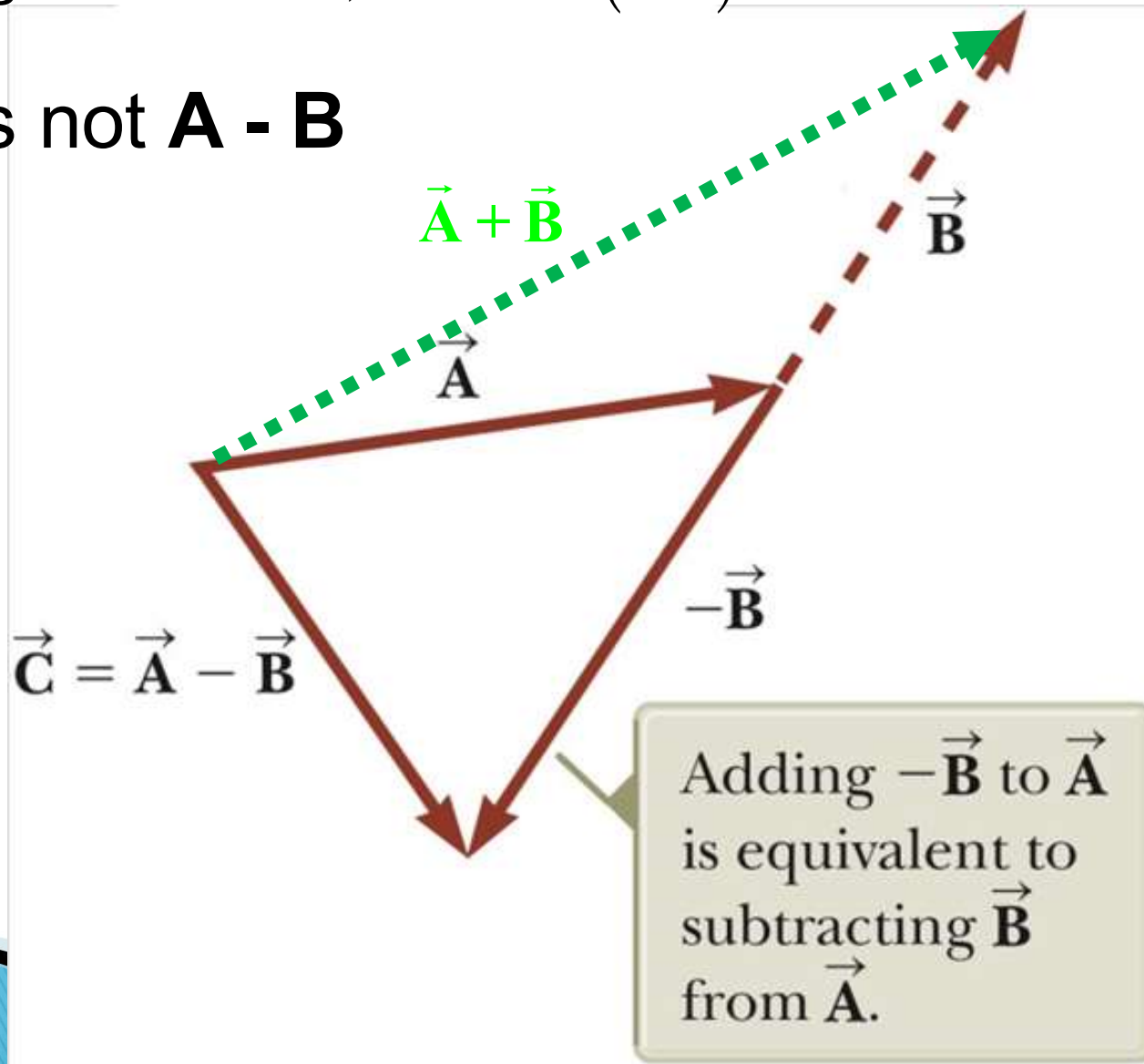


Vector addition–Geometric

- ▶ Vector subtraction
(Negative vector)

$$\vec{A} + (-\vec{B}) = \vec{C} = \vec{A} - \vec{B}$$

A + B is not A - B



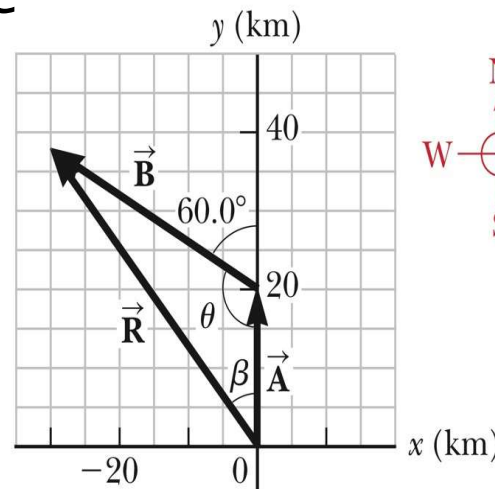
Example

The magnitudes of displacements $\vec{a}$ and $\vec{b}$ are 3 m and 4 m respectively, and $\vec{c} = \vec{a} + \vec{b}$. Considering the various orientations of $\vec{a}$ and $\vec{b}$, what is the (i) **maximum** and (ii) **minimum possible magnitude** for $\vec{c}$?

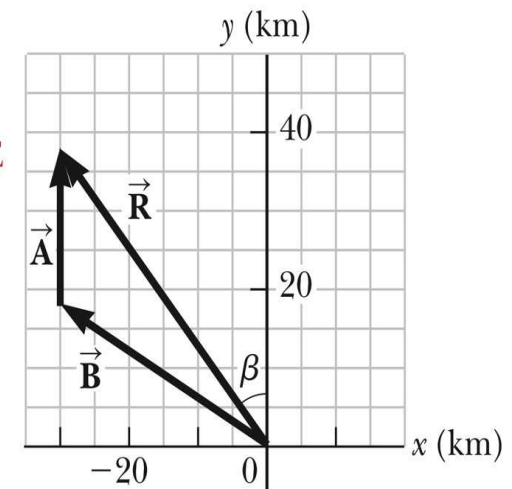
HOMEWORK Go through Example

A car travels 20.0 km due north and then 35.0 km in a direction 60.0° W of N. Find the magnitude & direction of the car's displacement. BY **GEOMETRICAL MEANS**

$X_f - X_i = \text{DEFINITIONN}$



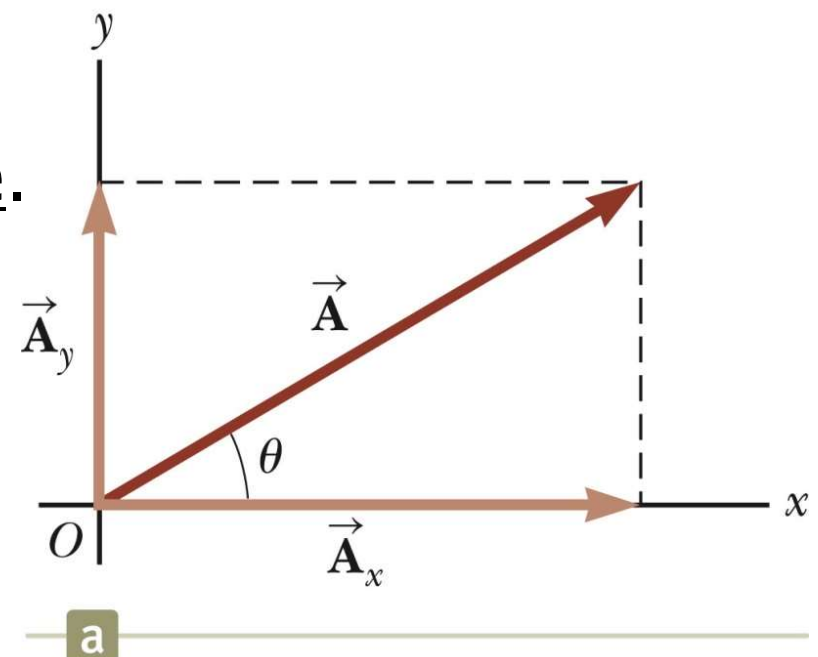
a



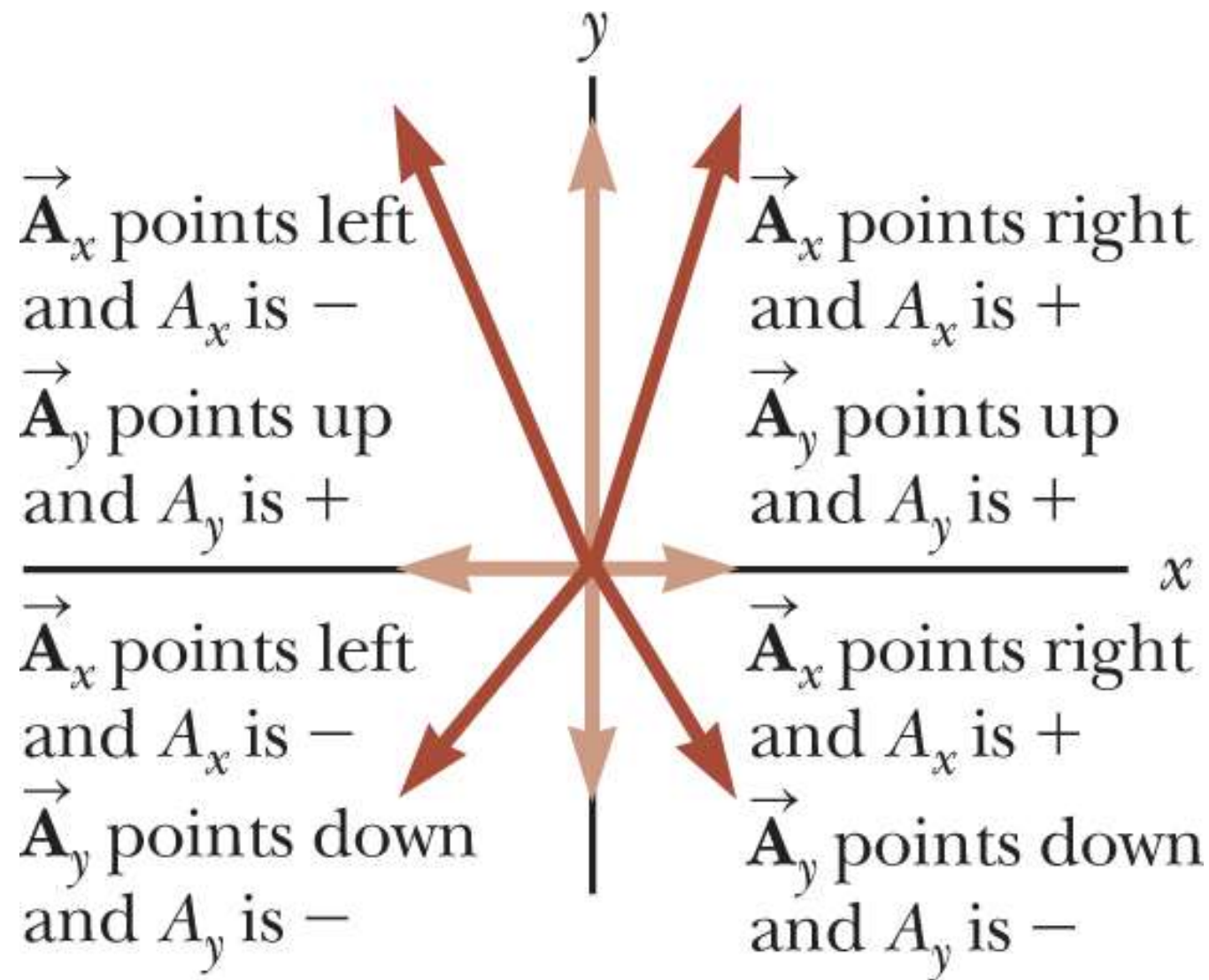
b

Components of vectors

- ▶ Graphical addition of vectors – inefficient
- ▶ **A component of a vector is a projection of the vector on an axis.**
- ▶ $\vec{A}_x$ is the component of vector $\vec{A}$ along the x -axis, and $\vec{A}_y$ is the component of the vector along the y -axis.
- ▶ Draw $\perp$ lines to axes.
- ▶ Resolve the vector.
- ▶ A_x, A_y both positive in this case.
- ▶ Generally, 3 components x -, y - and z -components.



Components of a Vector



Components of vectors

► Usually:

$$\cos \theta = \frac{adj}{hyp}$$

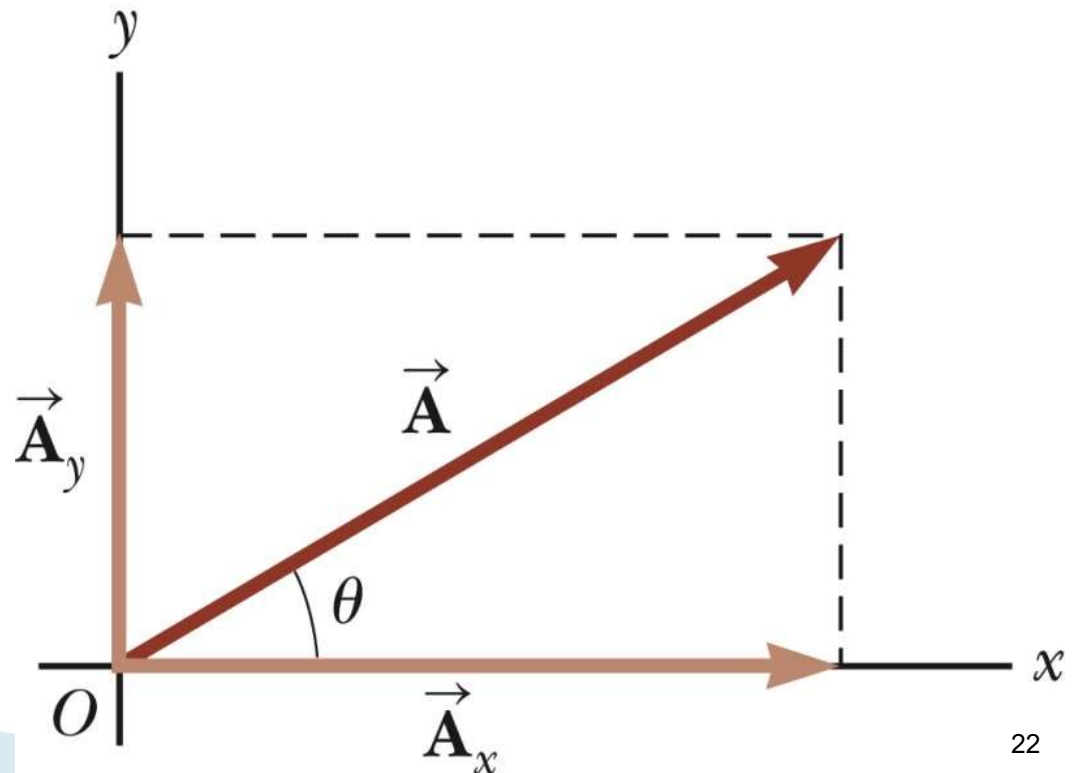
$$\text{then } A_x = A \cos \theta$$

$$\sin \theta = \frac{opp}{hyp}$$

$$\text{then } A_y = A \sin \theta$$

$$A = \sqrt{A_x^2 + A_y^2}$$

$$\tan \theta = \frac{opp}{adj} = \frac{A_y}{A_x}$$



Components of vectors

- Now, using the other angle – ϕ :

$$\cos \phi = \frac{adj}{hyp}$$

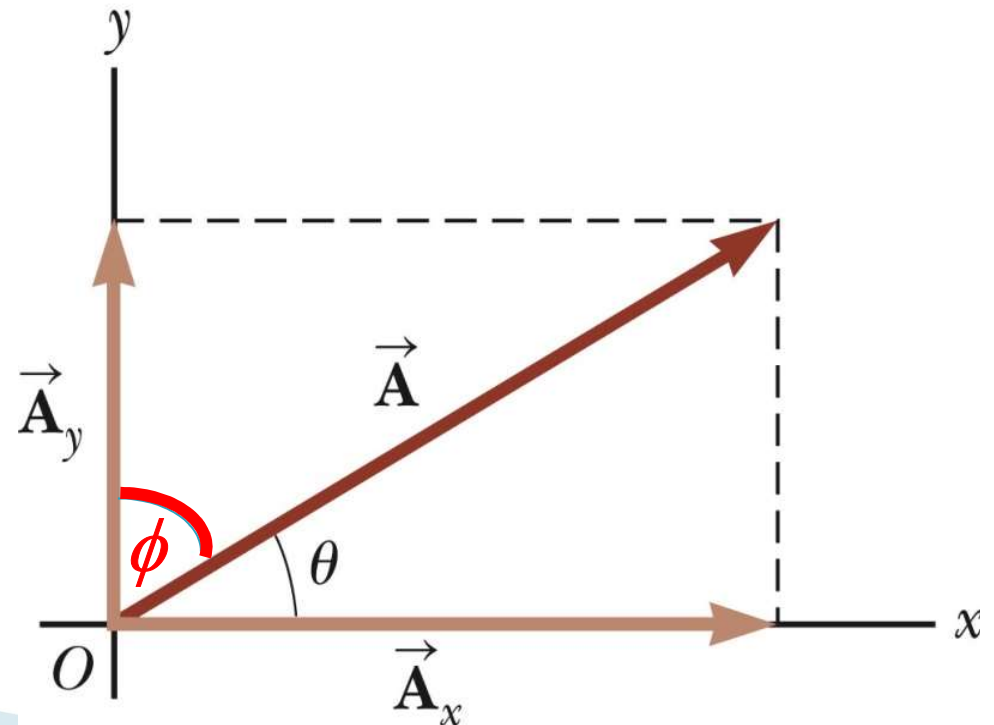
$$\text{then } A_y = A \cos \phi$$

$$\sin \phi = \frac{opp}{hyp}$$

$$\text{then } A_x = A \sin \phi$$

$$A = \sqrt{A_x^2 + A_y^2}$$

$$\tan \phi = \frac{A_x}{A_y}$$



Example,

A small plane leaves an airport on an overcast day and is later sighted 215 km away in a direction making an angle of 22° east of due north. How far east and north is the plane from the airport when it is sighted?

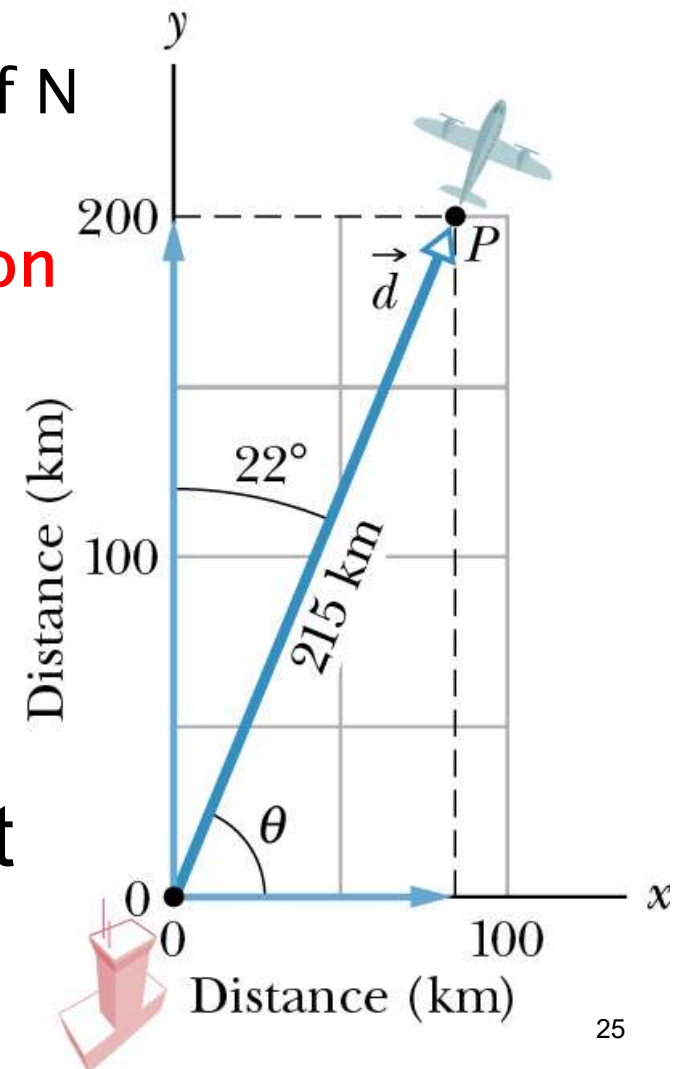


Example

- ▶ A small plane leaves an airport on an overcast day and is later sighted 215 km away in a direction making an angle of 22° east of due north. How far east and north is the plane from the airport when sighted?
- ▶ Given: distance: 215 km, angle 22° E of N
- ▶ Sketch
- ▶ **We can use θ or $\phi=22^\circ$ in computation**

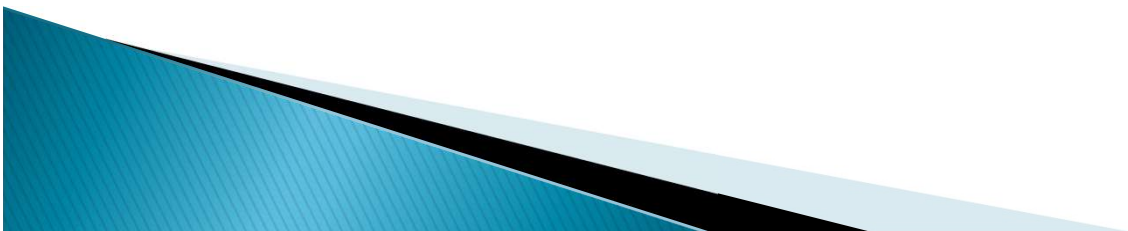
$$\begin{aligned}d_x &= d \sin \phi = 215 \sin 22^\circ \\ &= 81 \text{ km east of airport}\end{aligned}$$

$$\begin{aligned}d_y &= d \cos \phi = 215 \cos 22^\circ \\ &= 199 \approx 200 \text{ km north of airport}\end{aligned}$$



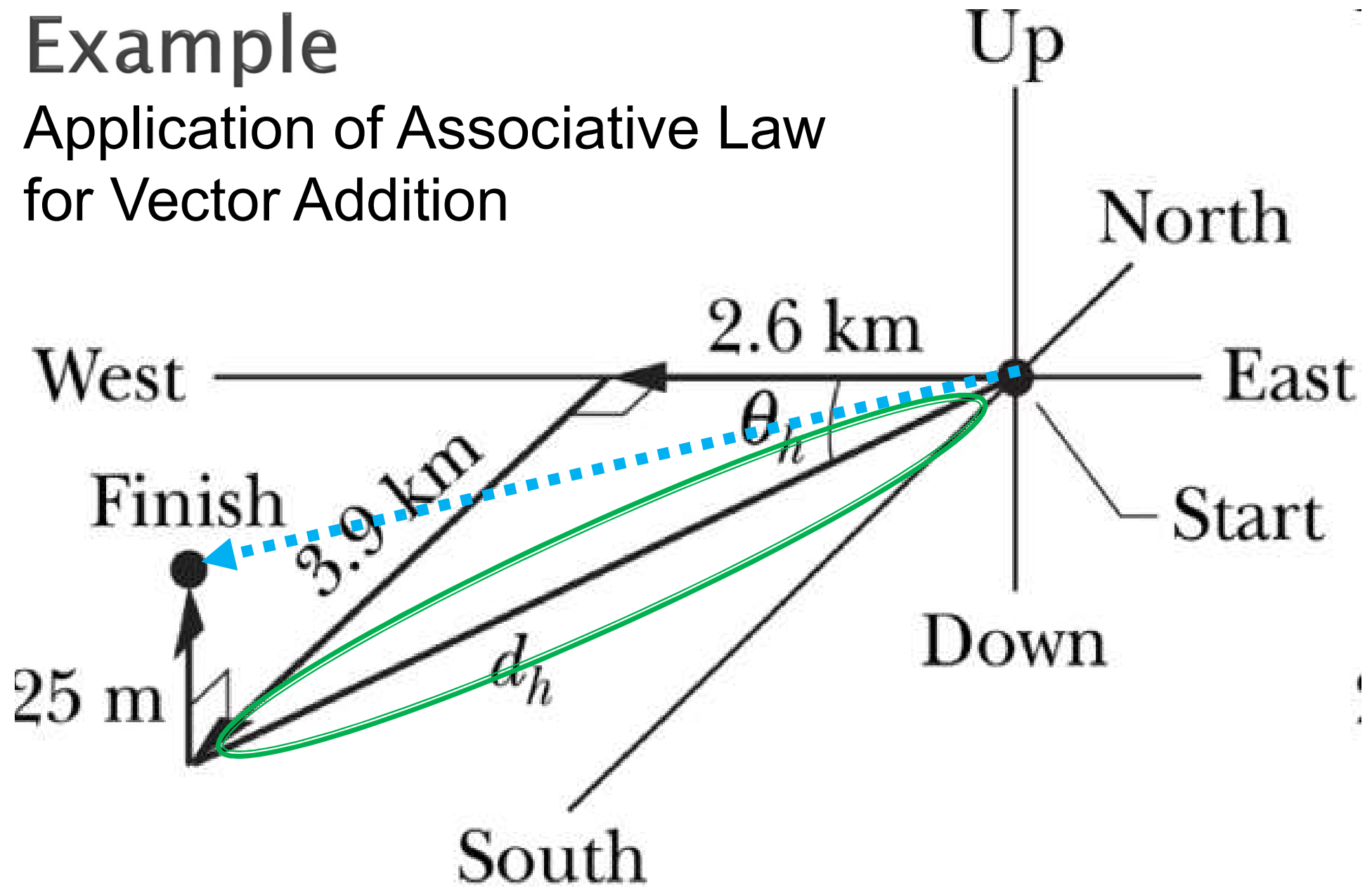
Example

- ▶ For 2 decades spelunking teams sought a connection between Flint Ridge cave system and Mammoth Cave (Kentucky, USA) When the connection was finally discovered, the combined system was declared the world's longest cave (>200 km) The team that found the connection had to crawl, climb and squirm through many passages, travelling a net 2.6 km westward, 3.9 km southward and 25m upward. What was their displacement from start to finish?



Example

Application of Associative Law
for Vector Addition



(a)

Example

- **3-dimensional:**

- **Horizontal components (W and S):**

$$d_h = \sqrt{(2.6\text{km})^2 + (3.9\text{km})^2} = 4.69 \text{ km}$$

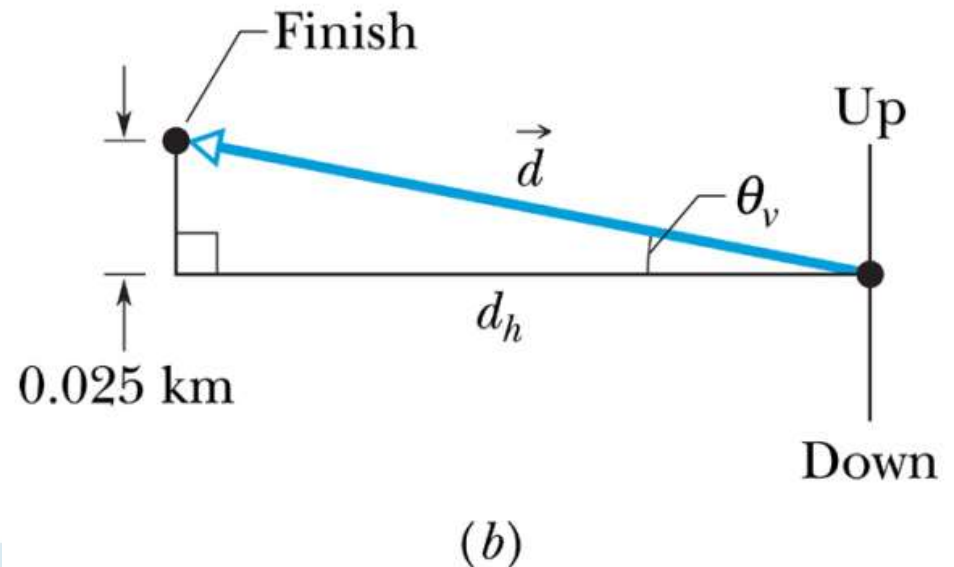
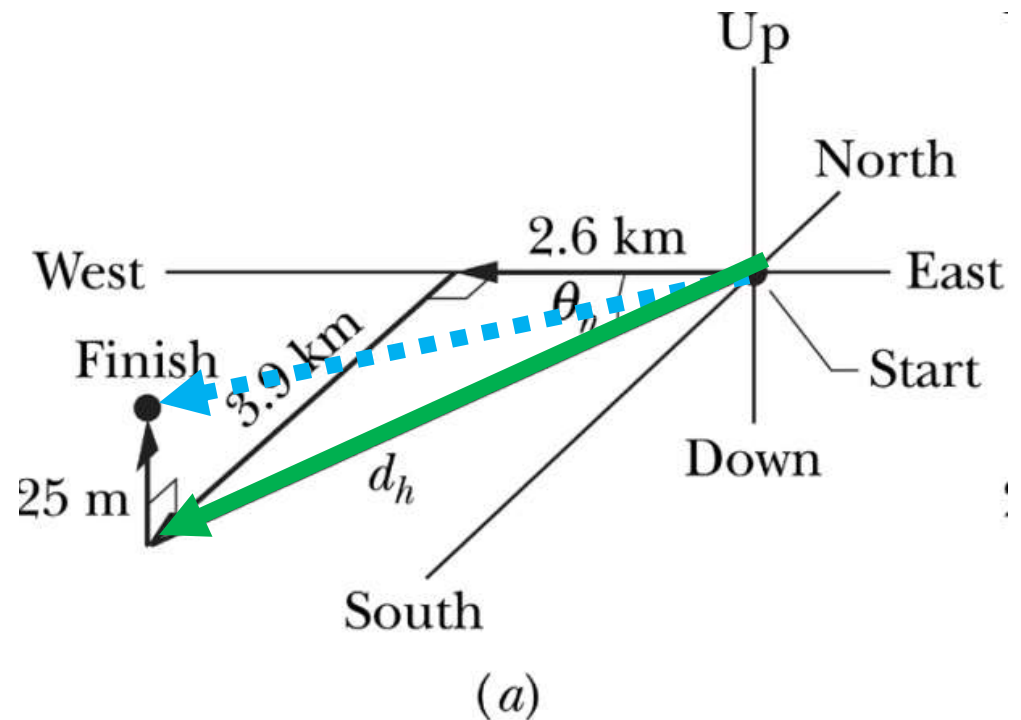
$$\theta_h = \tan^{-1}\left(\frac{3.9}{2.6}\right) = 56^\circ \text{ S of W}$$

- **Overall displacement:**

$$d_v = \sqrt{(4.69\text{km})^2 + (0.025\text{km})^2} \approx 4.7 \text{ km}$$

$$\theta_v = \tan^{-1}\left(\frac{0.025}{4.69}\right) = 0.3^\circ \text{ above horizontal}$$

Magnitude of 4.7 km at an angle 56° S of W and 0.3° above initial starting point.



Unit Vectors

- ▶ A unit vector has a magnitude of exactly 1 (one) and points in a particular direction.
- ▶ It is Dimensionless and unit-less.
- ▶ **Purpose is to specify direction.**
- ▶ Labelled in x, y, z direction as:

$\hat{\mathbf{i}}, \hat{\mathbf{j}}$ and $\hat{\mathbf{k}}$

- Right handed co-ordinate system

$$\vec{A} = A_x \hat{\mathbf{i}} + A_y \hat{\mathbf{j}}$$

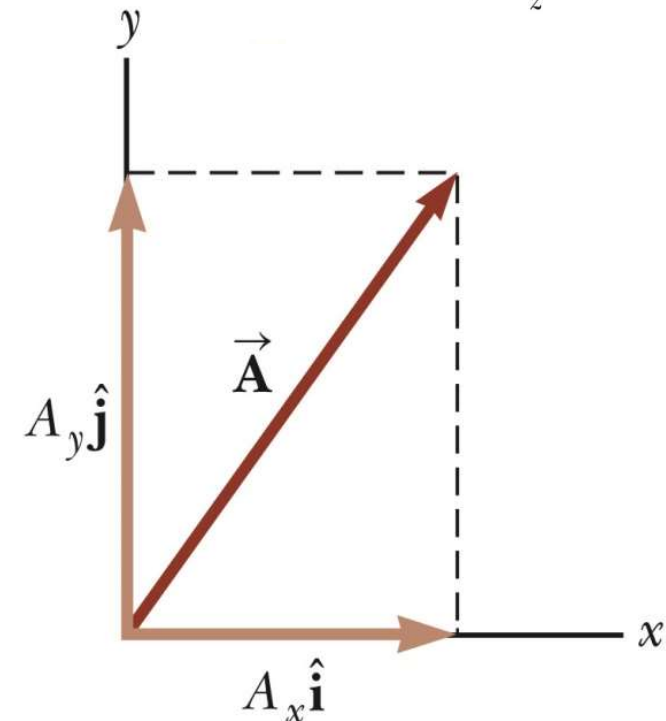
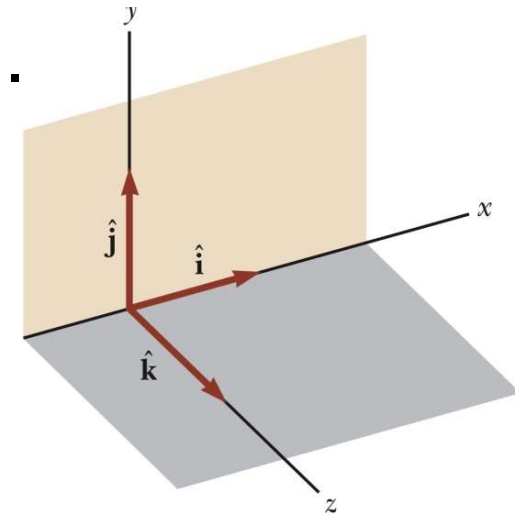
- Vector components (and units)

$$\vec{A}_x = A_x \hat{\mathbf{i}} \text{ and } \vec{A}_y = A_y \hat{\mathbf{j}}$$

- Scalar components (units)

$$A_x \text{ and } A_y$$

In general
$$\hat{\mathbf{A}} \equiv \frac{\vec{A}}{A} = \frac{A_x}{A} \hat{\mathbf{i}} + \frac{A_y}{A} \hat{\mathbf{j}} + \frac{A_z}{A} \hat{\mathbf{k}}$$



Adding Vectors by Components

- Consider the vectors

$$\vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

$$\vec{b} = b_x \hat{i} + b_y \hat{j} + b_z \hat{k}$$

$$\vec{r} = \vec{a} + \vec{b} \quad \text{Also, } \vec{r} - \vec{a} - \vec{b} = 0$$

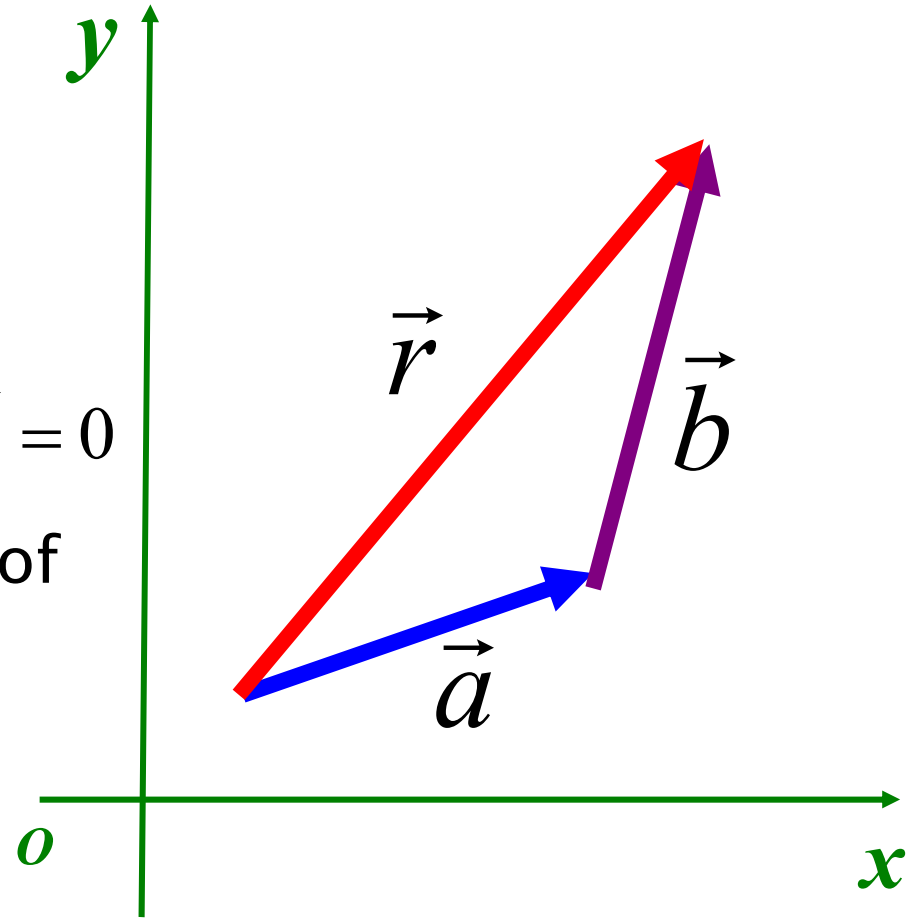
- Then Scalar Components of resultant vector r .

$$r_x = a_x + b_x$$

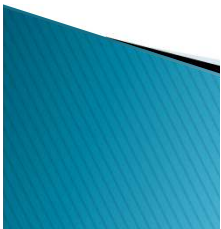
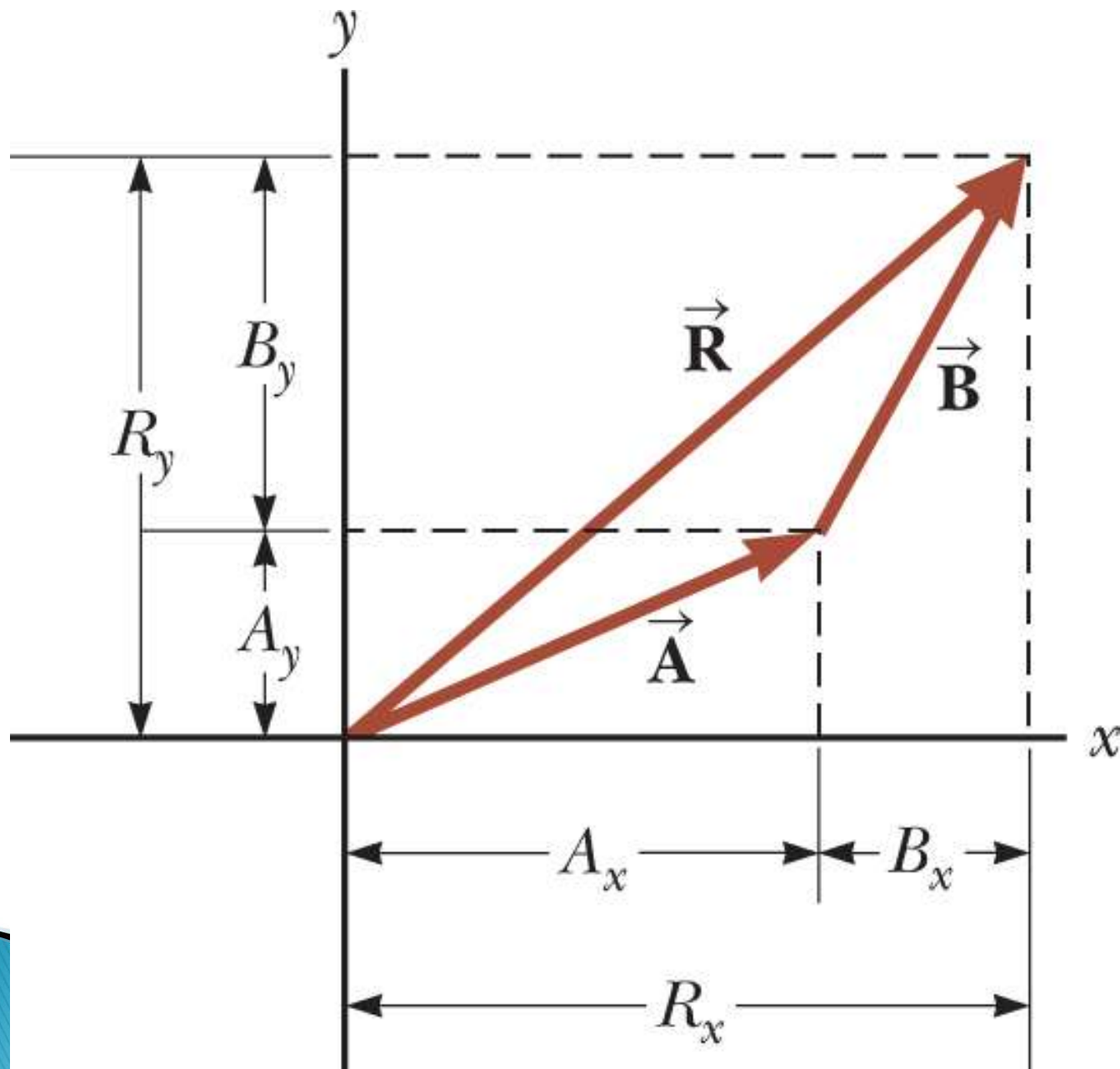
$$r_y = a_y + b_y$$

$$r_z = a_z + b_z$$

$$\begin{aligned} \vec{r} &= r_x \hat{i} + r_y \hat{j} + r_z \hat{k} \quad \text{resultant vector } \mathbf{r} \\ &= (a_x + b_x) \hat{i} + (a_y + b_y) \hat{j} + (a_z + b_z) \hat{k} \end{aligned}$$



Vector Addition using Components



Subtraction of Vectors

- Consider the vectors

$$\vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

$$\vec{b} = b_x \hat{i} + b_y \hat{j} + b_z \hat{k}$$

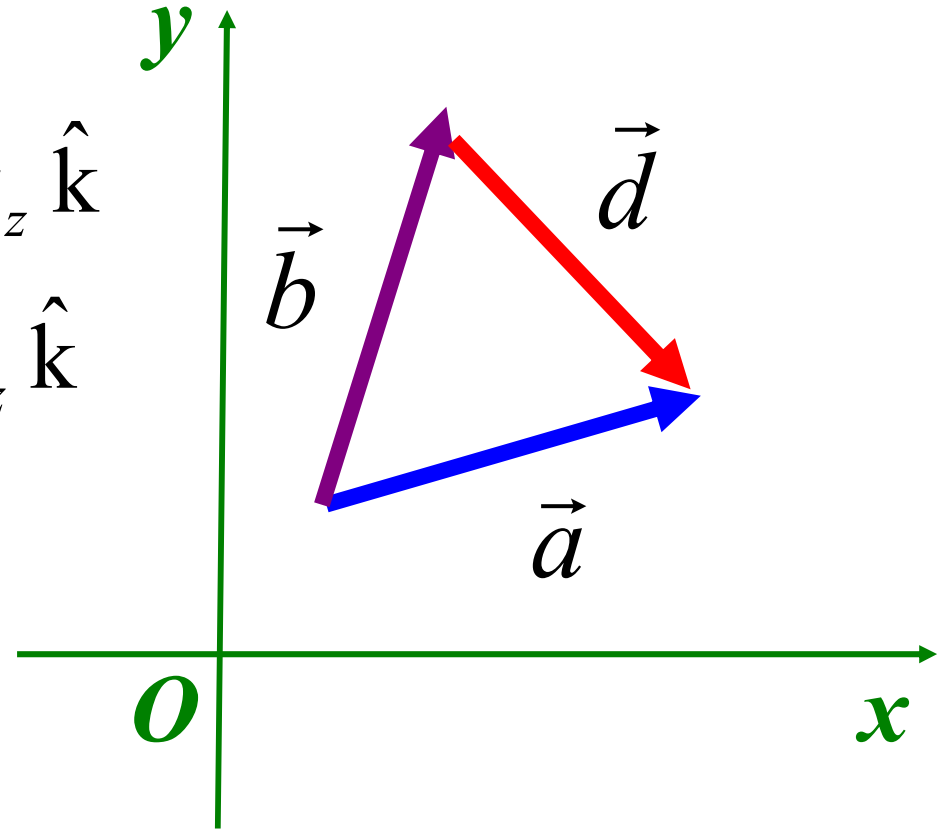
- Then $\vec{d} = \vec{a} - \vec{b}$

$$d_x = a_x - b_x$$

- and $d_y = a_y - b_y$

$$d_z = a_z - b_z$$

$$\vec{d} = d_x \hat{i} + d_y \hat{j} + d_z \hat{k}$$



Example

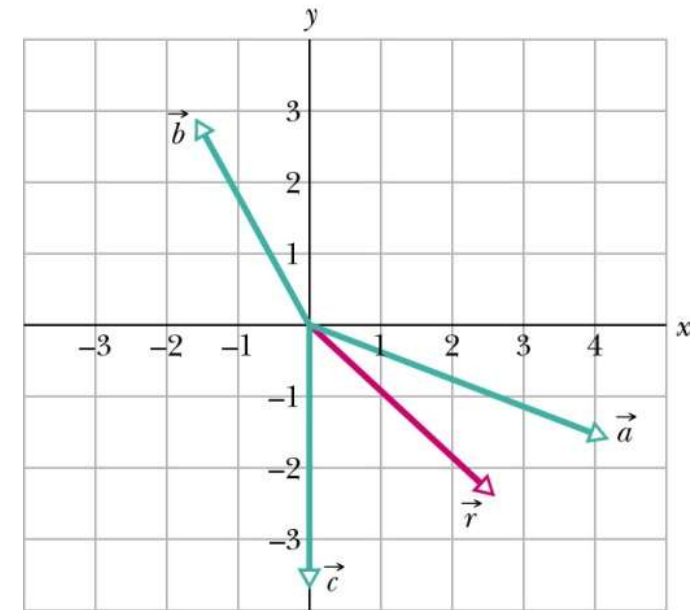
- Consider the three vectors as shown in the figure.

$$\vec{a} = (4.2 \text{ m})\hat{i} - (1.5 \text{ m})\hat{j}$$

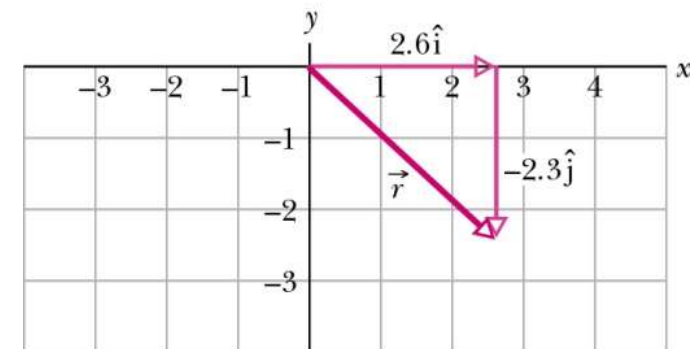
$$\vec{b} = (-1.6 \text{ m})\hat{i} + (2.9 \text{ m})\hat{j}$$

$$\vec{c} = (-3.7 \text{ m})\hat{j}$$

- What is their vector sum (shown, in purple)?

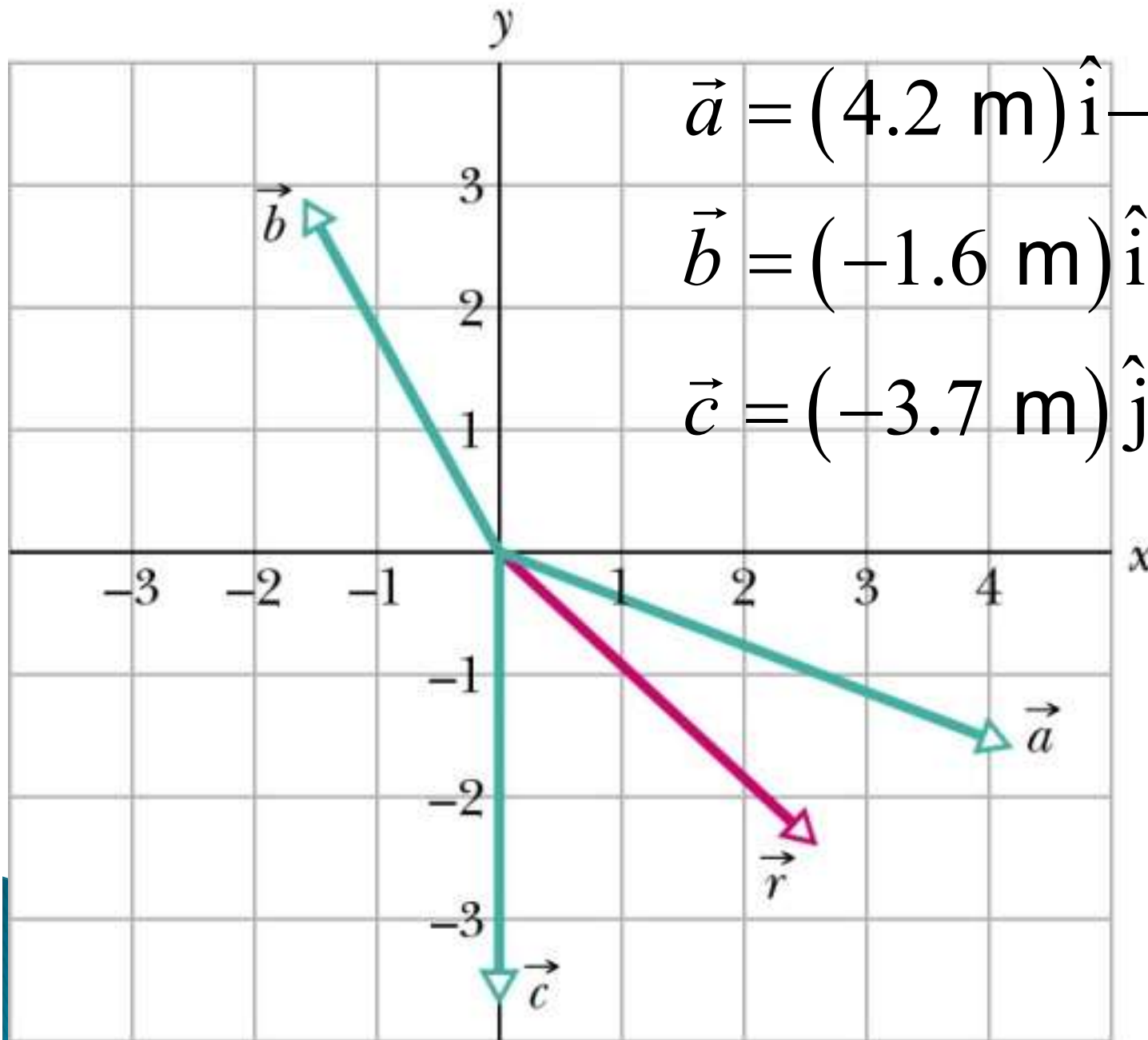


(a)



(b)

Example



$$\vec{a} = (4.2 \text{ m})\hat{i} - (1.5 \text{ m})\hat{j}$$

$$\vec{b} = (-1.6 \text{ m})\hat{i} + (2.9 \text{ m})\hat{j}$$

$$\vec{c} = (-3.7 \text{ m})\hat{j}$$

Example

- Consider the three vectors shown.

$$\vec{a} = (4.2 \text{ m})\hat{i} - (1.5 \text{ m})\hat{j}$$

$$\vec{b} = (-1.6 \text{ m})\hat{i} + (2.9 \text{ m})\hat{j}$$

$$\vec{c} = (0.0 \text{ m})\hat{i} + (-3.7 \text{ m})\hat{j}$$

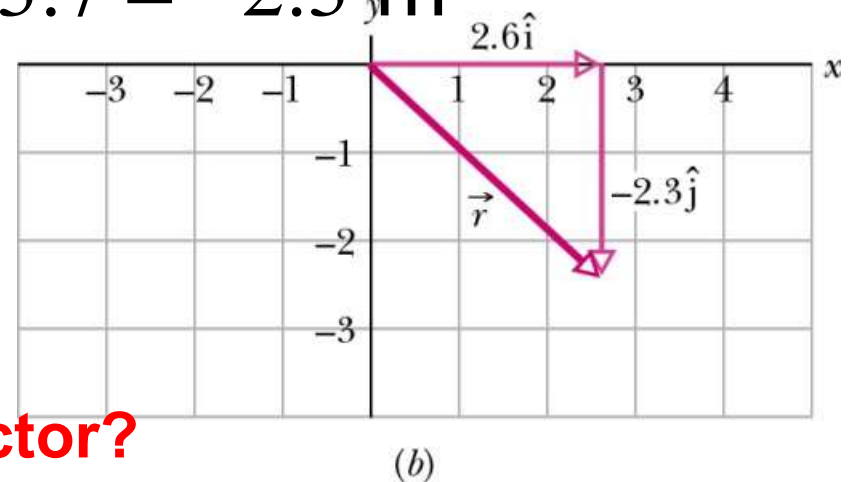
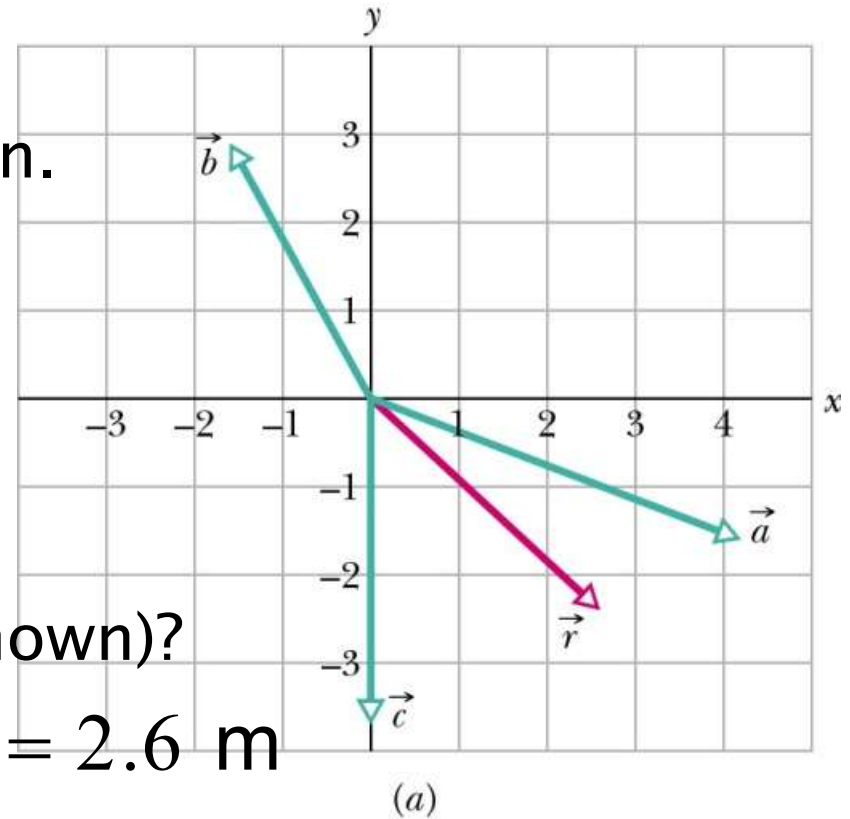
- What is their vector sum (also shown)?

$$r_x = a_x + b_x + c_x = 4.2 - 1.6 + 0 = 2.6 \text{ m}$$

$$r_y = a_y + b_y + c_y = -1.5 + 2.9 - 3.7 = -2.3 \text{ m}$$

$$r_z = a_z + b_z + c_z = 0 \text{ m}$$

$$\vec{r} = 2.6 \text{ m}\hat{i} - 2.3 \text{ m}\hat{j}$$

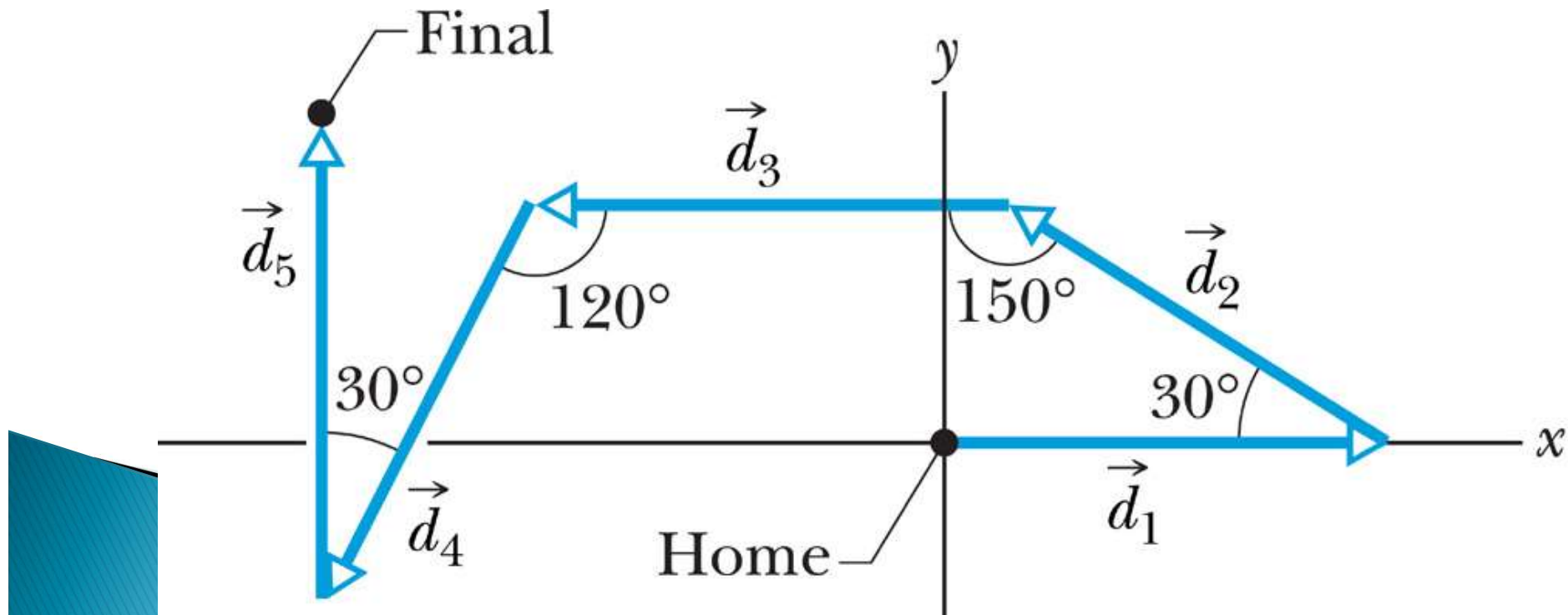


Magnitude?
Direction of vector?

Example

When an ant wants to return to its home next, it effectively sums its displacements along the axes of the system to calculate a vector that points directly home. Consider an ant making five runs of 6.0 cm each as shown in the figure.

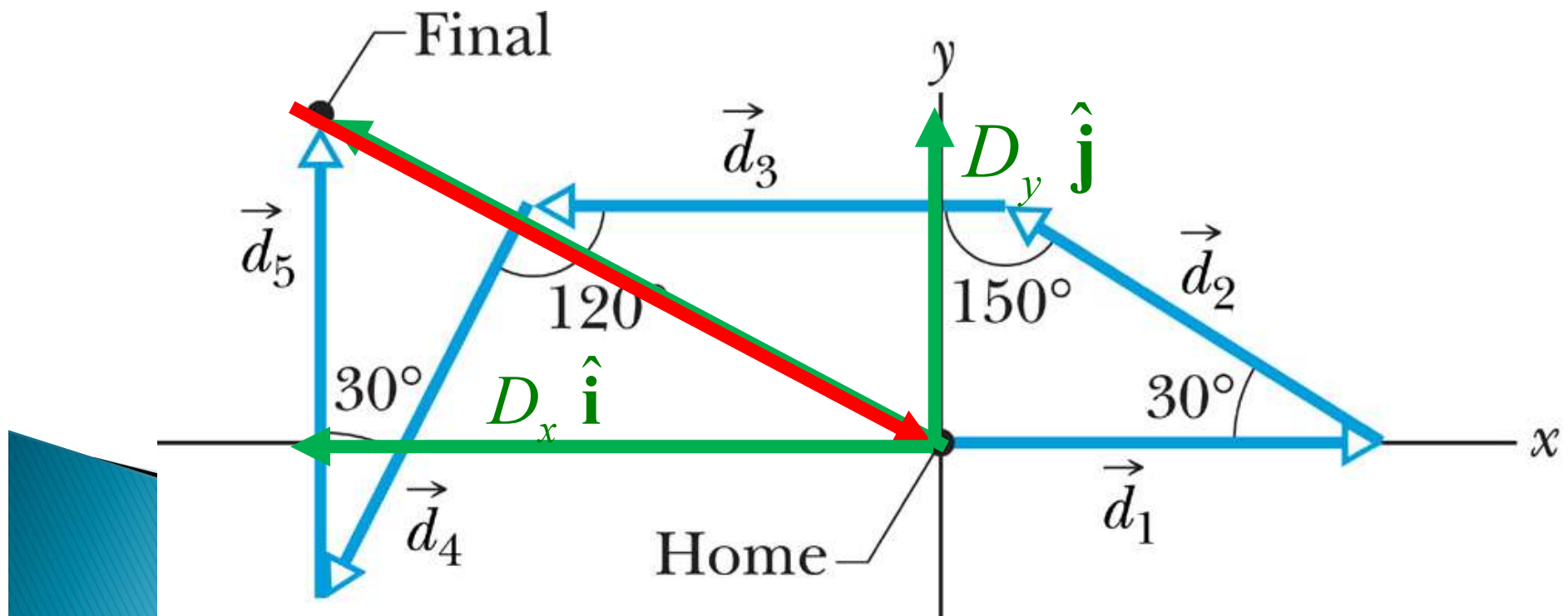
- What are the magnitude and direction of the ant's net displacement vector?
- What are the magnitude and direction of the homeward vector from the final position to the nest?



Example

$$\vec{D} = D_x \hat{\mathbf{i}} + D_y \hat{\mathbf{j}}$$

$$\vec{D}_x = \sum_{i=1}^5 d_i \hat{\mathbf{i}} = D_x \hat{\mathbf{i}} \quad \text{and} \quad \vec{D}_y = \sum_{i=1}^5 d_i \hat{\mathbf{j}} = D_y \hat{\mathbf{j}}$$



Multiplying Vectors

▶ Multiplying a vector by a scalar

$$s\mathbf{A}$$

- New vector is s times the magnitude of orig. vector
- Direction is same as original vector if s is positive
- Dividing vector by $s \Rightarrow$ multiply by $1/s$

▶ Multiplying a vector by a vector

$$\mathbf{A} \cdot \mathbf{B}$$

- **Two methods**
- One method produces a scalar – **scalar or dot product**

$$\mathbf{A} \times \mathbf{B}$$

- Other method produces a vector – **vector or cross product**

Scalar product

Two vectors $\vec{a}$ and $\vec{b}$

Scalar product or dot product

$$\vec{a} \cdot \vec{b} = ab \cos \phi$$

- a, b magnitudes of the vectors
- ϕ - angle between the directions of two vectors

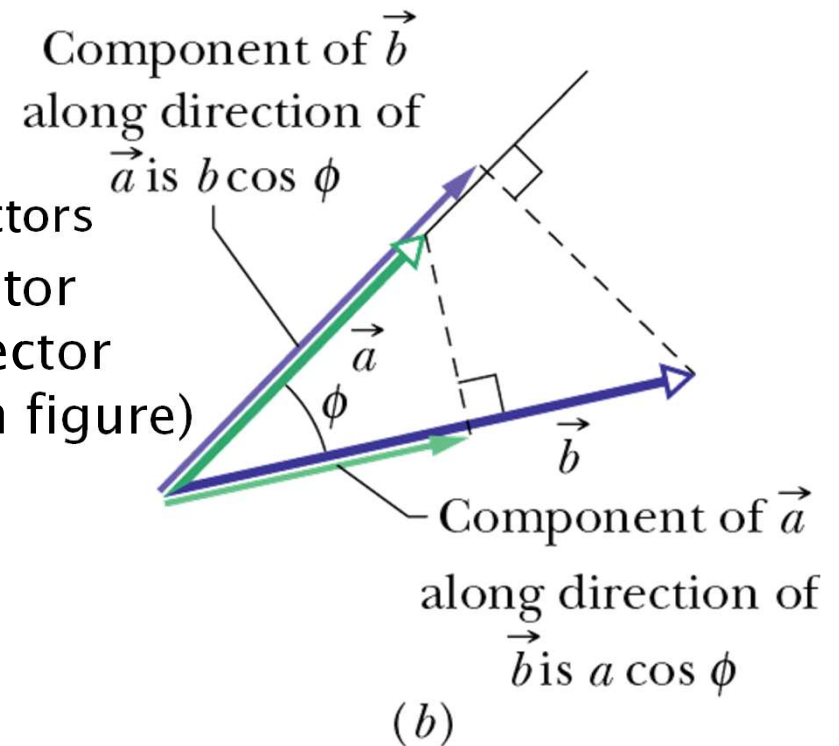
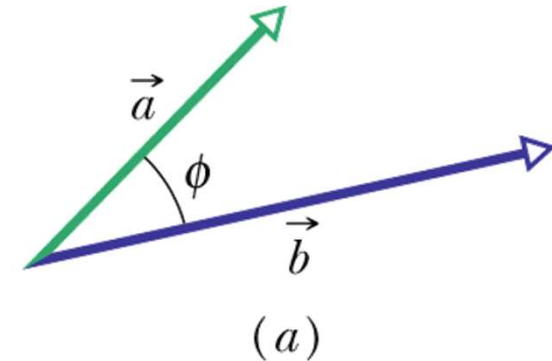
The product of the magnitude of one vector and the scalar component of the other vector along the direction of the first. (see (b) in figure)

If $\phi = 0^\circ$ then dot product is a max.

If $\phi = 90^\circ$ then dot product is 0.

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$$

$$\vec{a} \cdot \vec{b} = a_x b_x + a_y b_y + a_z b_z$$



Dot product PICKS out only components in same direction!

Scalar Product

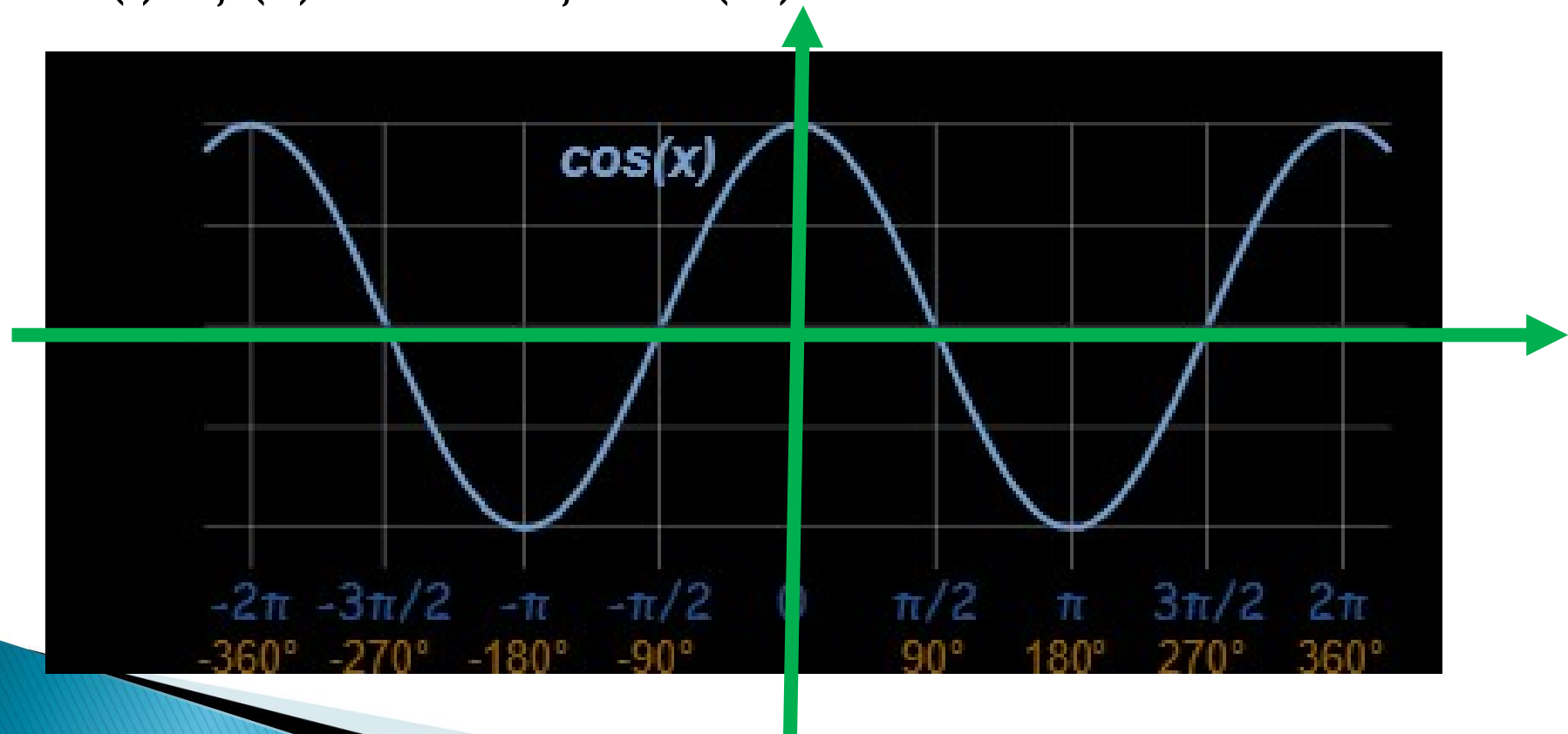
- Scalar product using unit vector notation

$$\begin{aligned}\vec{a} \cdot \vec{b} &= (a_x \hat{i} + a_y \hat{j} + a_z \hat{k}) \cdot (b_x \hat{i} + b_y \hat{j} + b_z \hat{k}) \\&= (\textcolor{red}{a_x \hat{i}} \cdot \textcolor{red}{b_x \hat{i}}) + (a_x \hat{i} \cdot b_y \hat{j}) + (a_x \hat{i} \cdot b_z \hat{k}) \\&\quad + (a_y \hat{j} \cdot b_x \hat{i}) + (\textcolor{blue}{a_y \hat{j}} \cdot \textcolor{blue}{b_y \hat{j}}) + (a_y \hat{j} \cdot b_z \hat{k}) \\&\quad + (a_z \hat{k} \cdot b_x \hat{i}) + (a_z \hat{k} \cdot b_y \hat{j}) + (\textcolor{violet}{a_z \hat{k}} \cdot \textcolor{violet}{b_z \hat{k}}) \\&= \textcolor{red}{a_x b_x} + \textcolor{blue}{a_y b_y} + \textcolor{violet}{a_z b_z}\end{aligned}$$

Dot product PICKS out only components in same direction!

Example

- ▶ Vectors $\vec{C}$ and $\vec{D}$ have magnitudes of 3 units and 4 units, respectively. What is the angle between the directions of the vectors if the dot product equals (i) 0, (ii) 12 units, and (iii) -12 units?



Examples

- ▶ What is the angle ϕ between

$$\vec{a} = 3.0\hat{i} - 4.0\hat{j} \text{ and } \vec{b} = -2.0\hat{i} + 3.0\hat{k} ?$$



Examples

- What is the angle ϕ between $\vec{a} = 3.0\hat{i} - 4.0\hat{j}$ and $\vec{b} = -2.0\hat{i} + 3.0\hat{k}$?

$$\vec{a} \cdot \vec{b} = ab \cos \phi \quad a = \sqrt{(3.0)^2 + (-4.0)^2} = 5.0$$

$$b = \sqrt{(-2.0)^2 + (3.0)^2} = 3.61$$

$$\begin{aligned}\vec{a} \cdot \vec{b} &= (3.0\hat{i} - 4.0\hat{j}) \cdot (-2.0\hat{i} + 3.0\hat{k}) \\ &= (3.0\hat{i}) \cdot (-2.0\hat{i}) + (3.0\hat{i}) \cdot (3.0\hat{k}) + (-4.0\hat{j}) \cdot (-2.0\hat{i}) + (-4.0\hat{j}) \cdot (3.0\hat{k}) \\ &= -6 + 0 + 0 + 0 = -6.0\end{aligned}$$

$$\vec{a} \cdot \vec{b} = ab \cos \phi$$

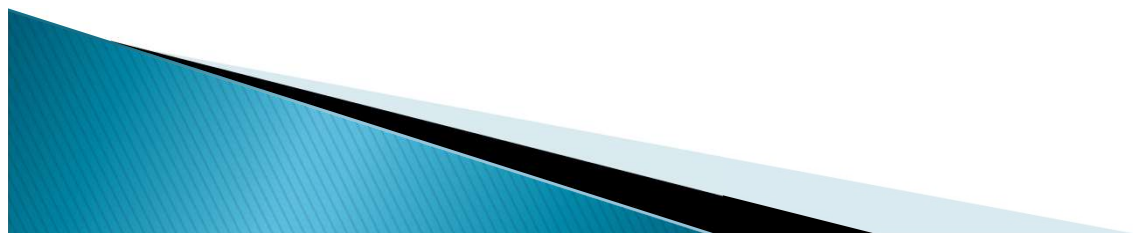
$$\cos \phi = \frac{\vec{a} \cdot \vec{b}}{ab} = \frac{-6.0}{(5.0)(3.61)} = 0.332$$

$$\phi = \cos^{-1}(0.332) = 109^\circ$$

If $\mathbf{B}$ is added to $\mathbf{C} = (3.0 \text{ m}) \hat{i} + (4.0 \text{ m}) \hat{j}$, the result is a vector in the positive direction of the y-axis, with a magnitude equal to that of $\mathbf{C}$. What is the magnitude of $\mathbf{B}$?

Thought Process:

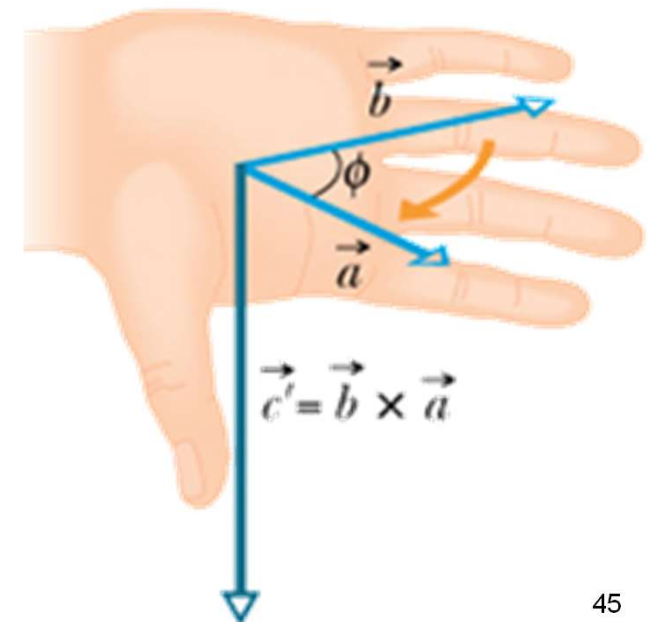
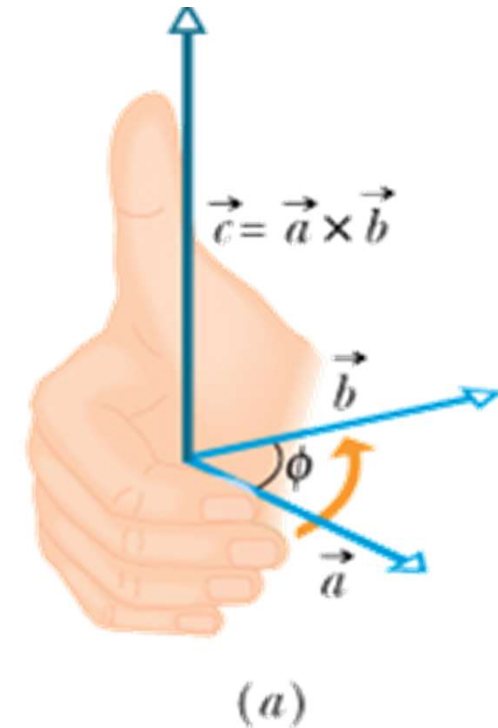
Sketch, Given info, Definitions, Simplification



Vector Product

- ▶ Consider two vectors $\vec{a}$ and $\vec{b}$
- ▶ Define Vector Product or Cross Product:
- ▶ $\vec{c} = \vec{a} \times \vec{b} = ab \sin \phi \hat{n}$
 - c – Magnitude of the third vector
 - a, b magnitudes of two initial vectors
 - ϕ – smaller of the two angle between the two vectors.
- ▶ If $\phi = 0^\circ$ then cross product is 0.
- ▶ If $\phi = 90^\circ$ then cross product is a max.
- ▶ Direction of $\vec{c}$ is $\perp$ to plane containing $\vec{a}$ and $\vec{b}$

▶ RIGHT-HAND RULE:



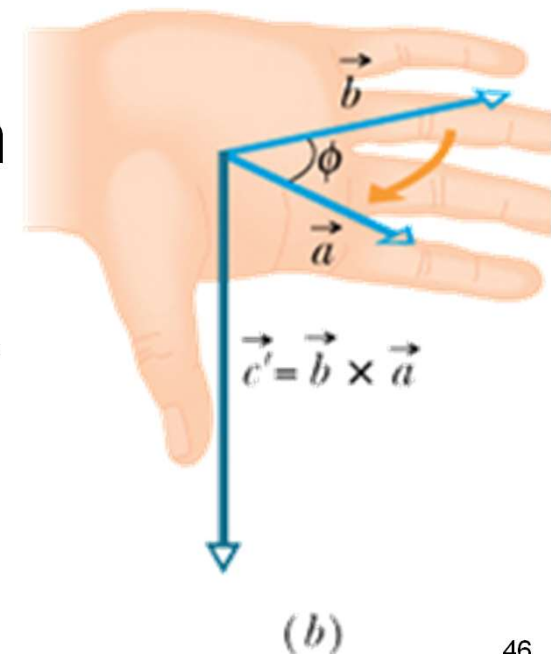
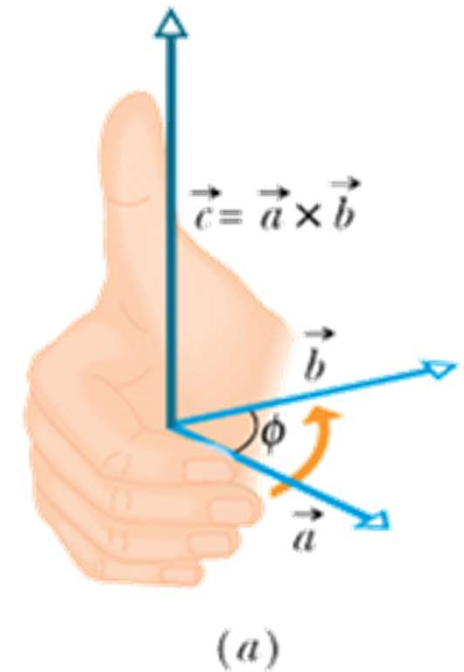
Vector Product – RH Rule

$$\vec{c} = \vec{a} \times \vec{b}$$

- ▶ **Place the vectors tail to tail.**
- ▶ Point the fingers of your right hand in the direction of the first vector.
- ▶ Curl your fingers, through the smallest angle between $\vec{a}$ and $\vec{b}$, from vector $\vec{a}$ to $\vec{b}$.
- ▶ Your outstretched thumb points in the direction of $\vec{c}$.
- ▶ Commutative law does NOT apply.

HomeWork: Prove this

$$\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$$



Vector Product

$$\vec{c} = \vec{a} \times \vec{b}$$

$$\begin{aligned}\vec{a} \times \vec{b} &= (a_x \hat{i} + a_y \hat{j} + a_z \hat{k}) \times (b_x \hat{i} + b_y \hat{j} + b_z \hat{k}) \\&= \underbrace{(a_x \hat{i} \times b_x \hat{i})} + (a_x \hat{i} \times b_y \hat{j}) + (a_x \hat{i} \times b_z \hat{k}) \\&\quad + (a_y \hat{j} \times b_x \hat{i}) + \underline{(a_y \hat{j} \times b_y \hat{j})} + (a_y \hat{j} \times b_z \hat{k}) \\&\quad + (a_z \hat{k} \times b_x \hat{i}) + (a_z \hat{k} \times b_y \hat{j}) + \underline{(a_z \hat{k} \times b_z \hat{k})} \\&= (a_y b_z - b_y a_z) \hat{i} + (a_z b_x - b_z a_x) \hat{j} + (a_x b_y - b_x a_y) \hat{k}\end{aligned}$$

► Determinant method

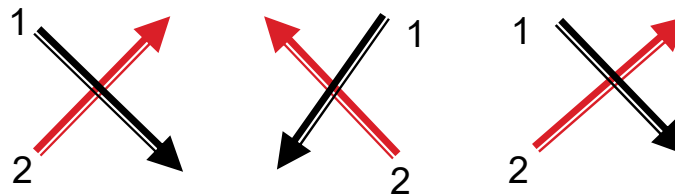


Vector Product

$$\vec{c} = \text{Det} \left| \vec{a} \times \vec{b} \right|$$

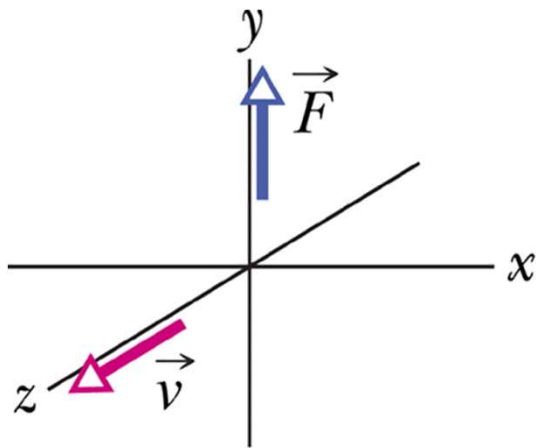
$$= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$$

$$\vec{a} \times \vec{b} = (a_y b_z - b_y a_z) \hat{\mathbf{i}} + (a_z b_x - b_z a_x) \hat{\mathbf{j}} + (a_x b_y - b_x a_y) \hat{\mathbf{k}}$$

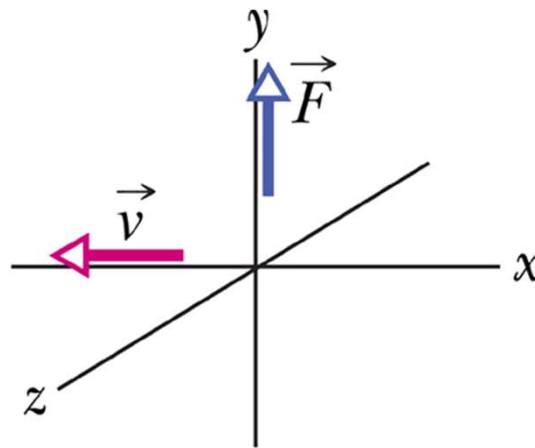


Example

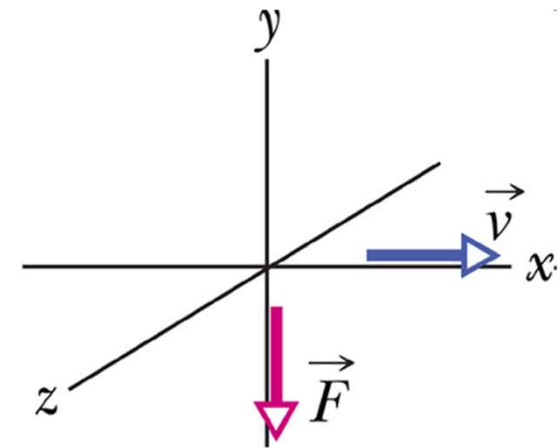
- ▶ If $\vec{F} = q(\vec{v} \times \vec{B})$ and $\vec{v}$ is perpendicular to $\vec{B}$, then what is the direction of $\vec{B}$ in each of the three situations below:



(1)



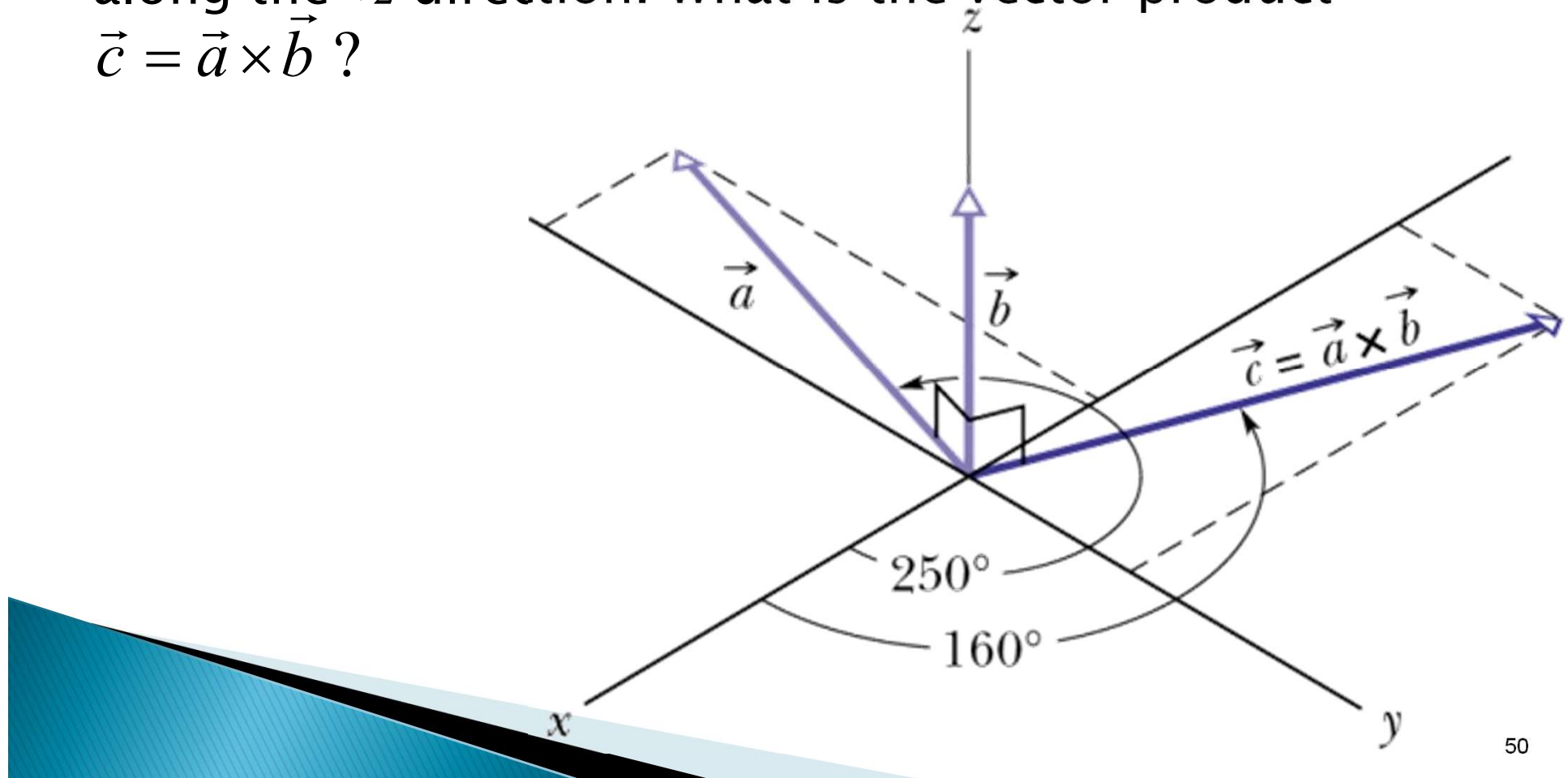
(2)



(3)

Example

- ▶ The figure shows vector $\vec{a}$ lying in the xy plane, with a magnitude of 18 units and pointing in the direction of 250° from the $+x$ -direction. Also, vector $\vec{b}$ has a magnitude of 12 units and points along the $+z$ direction. What is the vector product $\vec{c} = \vec{a} \times \vec{b}$?



Example

- ▶ The figure shows vector $\vec{a}$ lying in the xy plane, with a magnitude of 18 units and pointing in the direction of 250° from the $+x$ -direction. Also, vector $\vec{b}$ has a magnitude of 12 units and points along the $+z$ direction. What is the vector product?

$$\vec{c} = \vec{a} \times \vec{b}$$

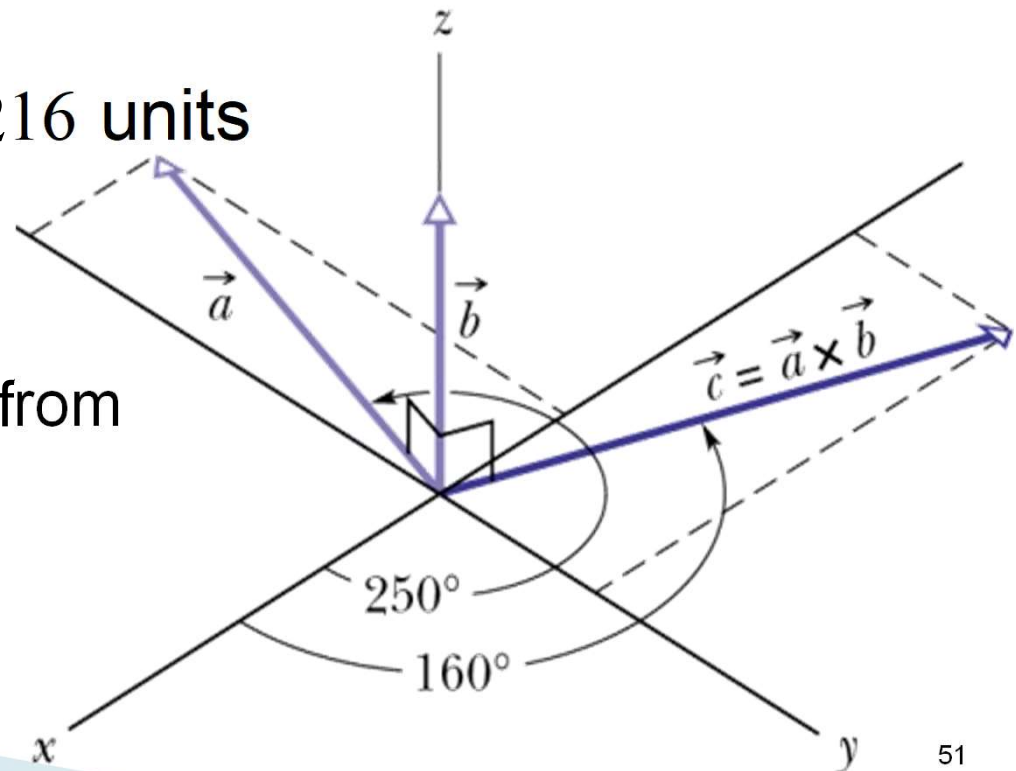
Magnitude:

$$c = ab \sin \phi = 18(12) \sin 90^\circ = 216 \text{ units}$$

Direction:

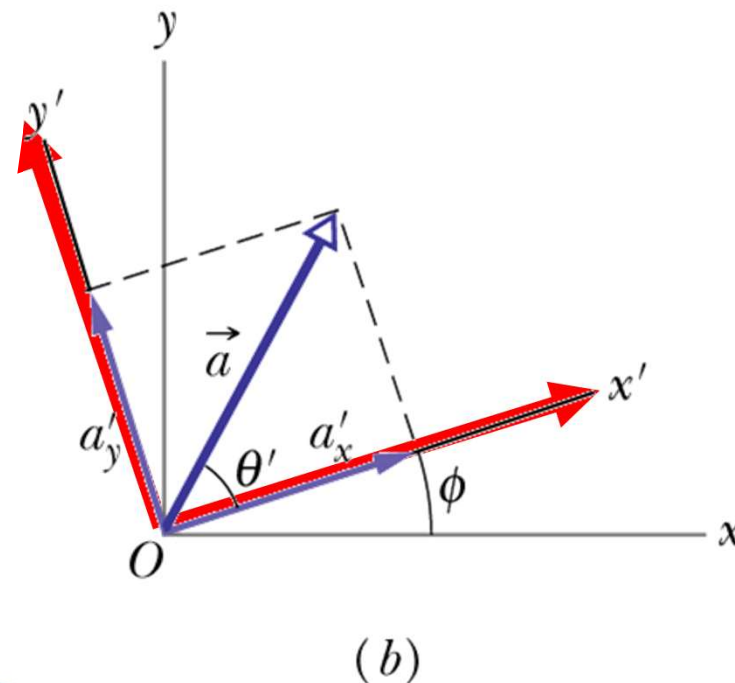
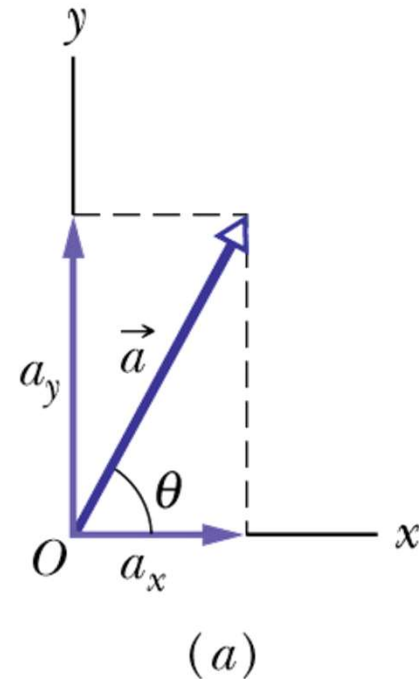
$\perp$ to $+z$ -axis.

$\perp$ to $\vec{a}$ so $250^\circ - 90^\circ = 160^\circ$ from $+x$ direction



Vectors and Physics

- ▶ Usually have shown axes; vectors parallel to the page edges.
- ▶ For Convenience reasons!
- ▶ Can rotate the axis and still have the “same” resultant vector. x' & y'
- ▶ Use this a lot in Physics particularly Mechanics.
- ▶ You CANNOT rotate the vectors themselves!!



Vectors and Physics

- ▶ Usually have shown axes; vectors parallel to the page edges.
- ▶ No particular reason for this.
- ▶ Can rotate the axis and still have the “same” resultant vector.
- ▶ Use this a lot in Physics particularly Mechanics.
- ▶ You CANNOT rotate the vectors themselves!!

